BEYOND BROWN'S RADIAL TESTS: ADMISSIBILITY, PARABOLICITY, AND A DOUBLING–POINCARÉ PRIOR THAT NEITHER TEST DECIDES

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Abstract. For the estimation of a normal mean in $\mathbb{R}^m$ under quadratic loss, Brown (1971) proved that a generalized Bayes estimator δ_F with bounded drift is admissible if and only if a diffusion attached to the marginal f^* of the prior F is recurrent, and gave two radial tests: a sphere-average test for recurrence and a ray test for transience. He observed that both tests can be silent. We read his theory as the potential theory of the weighted Euclidean space $(\mathbb{R}^m, f^*\,\mathrm{d}x)$ and ask what the modern theory of recurrence on weighted manifolds adds inside the zone where neither test applies. Three results. (A) If $f^*\,\mathrm{d}x$ is doubling and supports a scale-invariant Poincaré inequality, the sphere test is already complete: recurrence, Brown's sphere test and the ball-volume test are equivalent, so geometric criteria can add decidable cases only on the inadmissible side. (B) They do add one. An explicit prior on $\mathbb{R}^3$ whose marginal is small along a lattice of circles at dyadic scales with spacing equal to the square root of the scale is doubling and Poincaré, silences both of Brown's tests, and is transient by the ball-volume test; its generalized Bayes estimator is inadmissible. (C) Outside the doubling–Poincaré class the zone is two-sided: the same circles placed on one surface per scale give an admissible estimator, decided by a flux bound along a non-spherical exhaustion, and a logarithmic product prior gives an inadmissible one, decided by product heat kernels. A comparison lemma shows that any positive weight with Lipschitz logarithm is, up to constants, the marginal of a prior, so that explicit weights may be used throughout.

1. Introduction

Let $X \sim N_m(\theta, I_m)$ and consider estimators of θ under quadratic loss. For a σ-finite measure F on $\mathbb{R}^m$ with finite Gaussian marginal $f^* = \varphi * F$, where φ is the standard normal density, the generalized Bayes estimator is

$$\delta_F(x) = x + \nabla \log f^*(x).$$

Brown [5] showed that every admissible estimator is of this form and characterized admissibility by the behaviour of a diffusion: let $\{Z_t\}$ be the diffusion on $\mathbb{R}^m$ with generator $\Delta + \nabla \log f^* \cdot \nabla$, the weighted Laplacian of the measure $\mu = f^*\,\mathrm{d}x$. Then δ_F is inadmissible if $\{Z_t\}$ is transient, and admissible if $\{Z_t\}$ is recurrent and the drift $\nabla \log f^*$ is bounded. Since the recurrence of $\{Z_t\}$ is the *parabolicity* of the weighted Euclidean space $W_F = (\mathbb{R}^m, \delta_{ij}, f^*\,\mathrm{d}x)$ (Figure 1), Brown's theorem places the admissibility problem inside the potential theory of weighted manifolds, a subject that has developed independently since [27, 14, 12]. The statistical literature has long used the Dirichlet-form formulation of Brown's criterion [19, 18, 9, 37]; the geometric reading, and the tests it brings, are the subject of this note.

Brown gave two tests that depend only on radial information. Write $\bar{f}(r)$ for the mean of f^* over the sphere $\{|x| = r\}$. His sphere test says that $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r)) = \infty$ implies recurrence; his ray test says that if the ray integrals $\int_1^\infty \mathrm{d}r/(r^{m-1}f^*(r\varphi))$ are uniformly bounded on a set of directions φ of positive measure, then δ_F is inadmissible. He remarked [5, p. 900] that there are marginals on which neither test applies. We call this set of marginals *Brown's undecided zone*.

The geometric literature knows the sphere test as the sphere-area parabolicity test [12, Thm. 7.5], and knows further tests with no statistical counterpart: the ball-volume test of Karp, Varopoulos

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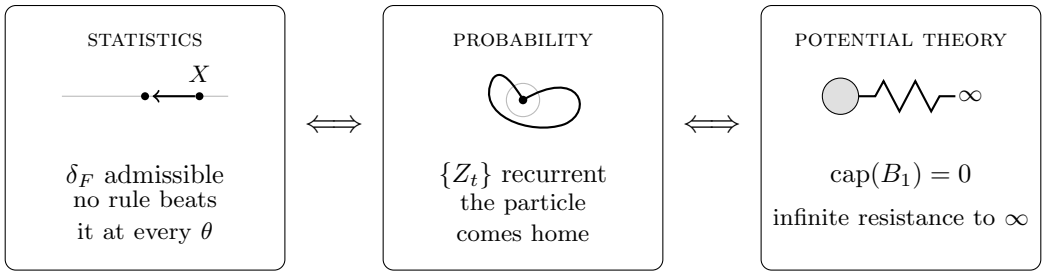


FIGURE 1. Brown's dictionary: one fact in three languages (Theorem 2.1). A generalized Bayes estimator with bounded drift is admissible exactly when its weighted Brownian motion is recurrent, exactly when the unit ball has zero capacity, that is, infinite resistance to infinity. The negations are equivalent too: transient, inadmissible, positive capacity, current escapes.

and Grigor'yan, which is an equivalence when the measure is doubling and supports a Poincaré inequality [12, Thm. 11.1], [36, Cor. 4.12], and the flux test along an arbitrary exhaustion [12, Thm. 7.6]. Our question is what these add inside Brown's undecided zone. The answer has three parts.

(A) *Inside the doubling–Poincaré class, Brown's sphere test is complete* (Theorem 3.1). If $f^*\,\mathrm{d}x$ is doubling and supports a scale-invariant Poincaré inequality, then recurrence, the divergence of the sphere integral, and the divergence of the volume integral $\int^\infty r\,\mathrm{d}r/\mu(B_r)$ are equivalent. So for such priors with bounded drift, δ_F is admissible if and only if Brown's 1971 test says so, and geometric criteria can only decide cases on the inadmissible side.

(B) *They do decide one* (Theorem 4.1). We construct a prior on $\mathbb{R}^3$ whose marginal is doubling and Poincaré, on which both of Brown's tests are silent (every ray integral diverges, the sphere integral converges), and whose volume grows like r^3; the ball-volume test shows transience and δ_F is inadmissible. The construction places latitude circles on spheres of radius $\asymp 2^k$ in the shell $2^k \le |x| < 2^{k+1}$, at mutual distance $\asymp 2^{k/2}$, so that their angular positions cover every direction while no coherent surface of small weight forms.

(C) *The zone is two-sided* (Section 5). Placing the same circles on one surface per shell produces a prior on which both tests are silent and which is nevertheless recurrent, hence admissible; the proof is a flux bound along the surfaces. A logarithmic product prior is transient, hence inadmissible, by product heat kernels. Neither prior is doubling and Poincaré, and Theorem 3.1 says why the first cannot be.

A comparison lemma (Lemma 2.3) shows that any positive weight w with Lipschitz logarithm satisfies $c\,w \le \varphi * w \le C\,w$ with bounded drift, so that every explicit weight below is, up to constants, the marginal of the prior $F = w\,\mathrm{d}\theta$. All statements about recurrence, doubling, Poincaré inequalities and Brown's tests are stable under such comparability.

The examples of Section 5 connect to the question of Johnstone and Lalley [19], whether any generalized Bayes estimator that is not proper Bayes is immune to the Stein effect under products; their theory decides the product example and our construction in Theorem 4.1 is the non-product counterpart. The same dichotomy between parabolic and non-parabolic geometry governs Komaki's theory of shrinkage priors for predictive densities, where a prior asymptotically dominating the Jeffreys prior exists if and only if the Fisher manifold of the model is non-parabolic [24]; Appendix A records that parallel and the identity that expresses Komaki's risk reduction through Perelman's weighted scalar curvature.

Organization. Section 2 fixes notation, restates Brown's theorems in geometric language, proves the comparison lemma, and records the volume criterion. Section 3 proves Theorem 3.1 and locates

the undecided zone through the spherical A_2 ratio. Section 4 constructs the prior of Theorem 4.1 and proves its properties. Section 5 gives the two further examples. Section 6 reports numerical companions to the proofs, and Section 7 discusses what the geometric criteria add and what remains open. Appendix B records which statements of this note have been checked formally in the proof assistant Lean, and under which cited premises.

2. Preliminaries

2.1. **Notation.** Throughout, $m \geq 2$, $B(x,\rho)$ is the open Euclidean ball, $B_r = B(0,r)$, and ω_{m-1} the area of the unit sphere. For a positive locally integrable weight w on $\mathbb{R}^m$ we write $\mu = w\,\mathrm{d}x$ and $V(r) = \mu(B_r)$. For $r \geq 1$ let $\bar{f}(r)$ and $A(r)$ be the means of w and of $1/w$ over the sphere $\{|x| = r\}$, and $Q(r) = \bar{f}(r)A(r) \geq 1$ (Jensen) the *spherical A_2 ratio*. The measure μ is *doubling* if $\mu(B(x,2\rho)) \leq D\,\mu(B(x,\rho))$ for all balls, and supports the *scale-invariant Poincaré inequality* if

$$\int_B |u - u_B|^2\,\mathrm{d}\mu \;\leq\; C_P\,\rho^2 \int_B |\nabla u|^2\,\mathrm{d}\mu \qquad \text{for every ball } B \text{ of radius } \rho \text{ and every } u \in C^1(\bar{B}), \tag{1}$$

where u_B is the μ-mean of u on B. We say that μ (or w) is $D{+}P$ if both hold. The best constant in (1) on a given ball is $1/(\lambda_1(B)\rho^2)$, where $\lambda_1(B)$ is the first nonzero Neumann eigenvalue of the weighted Laplacian $-w^{-1}\operatorname{div}(w\nabla\cdot)$ on B; this is how the inequality is tested numerically in Section 6.

The weighted space $(\mathbb{R}^m, \delta_{ij}, \mu)$ carries the Dirichlet form $\mathcal{E}(u) = \int |\nabla u|^2\,\mathrm{d}\mu$ and the weighted Brownian motion with generator $\Delta + \nabla \log w \cdot \nabla$. It is *parabolic* if the capacity of some (equivalently every) ball vanishes, $\operatorname{cap}(B_1) = \inf\{\mathcal{E}(j) : j \in C_c^\infty,\ j \geq 1 \text{ on } B_1\} = 0$, equivalently if the weighted Brownian motion is recurrent; otherwise it is *non-parabolic* or *transient*. For Riemannian manifolds this equivalence is [12, Thm. 5.1]. For the weighted space with w continuous and positive it follows from the characterization of recurrence of a Dirichlet form by a sequence $u_n \to 1$ almost everywhere with $\mathcal{E}(u_n) \to 0$ [11, Thm. 1.6.3]: such a sequence yields test functions for $\operatorname{cap}(B_1)$ by truncation and a cut-off, and conversely test functions j_n with $\mathcal{E}(j_n) \to 0$, truncated to $[0,1]$, tend to 1 in L^2 of every ball by the Poincaré inequality of that ball, which holds with a constant depending on the ball because w is bounded above and below there.

2.2. **Brown's theorems.** We restate the results of [5] that we use, in the notation above with $w = f^*$.

Theorem 2.1 (Brown [5])**.** *Let F be a σ-finite measure on $\mathbb{R}^m$ with $f^* = \varphi * F < \infty$, and let $\{Z_t\}$ be the weighted Brownian motion of $(\mathbb{R}^m, f^*\,\mathrm{d}x)$.*

(a) (Thm. 4.3.1) *$\{Z_t\}$ is transient if and only if $\operatorname{cap}(B_1) > 0$ in $(\mathbb{R}^m, f^*\,\mathrm{d}x)$.*

(b) (Thm. 5.1.1) *If $\{Z_t\}$ is transient, δ_F is inadmissible. If $\{Z_t\}$ is recurrent and $\sup|\nabla \log f^*| < \infty$, δ_F is admissible.*

(c) (Thm. 6.3.1) *If F is spherically symmetric and $\nabla \log f^*$ is bounded, δ_F is inadmissible if and only if*

$$\int_1^\infty \frac{\mathrm{d}r}{r^{m-1} f^*(r)} < \infty;$$

the boundedness is used only for the admissible direction, as in (b).

(d) (Thm. 6.4.3, sphere test) *If*

$$\int_1^\infty \frac{\mathrm{d}r}{r^{m-1} \bar{f}(r)} = \infty,$$

then $\{Z_t\}$ is recurrent; hence δ_F is admissible if moreover $\nabla \log f^*$ is bounded.

(e) (Thm. 6.4.4, ray test) *If a set Ω of directions of positive surface measure satisfies*

$$\sup_{\varphi \in \Omega} \int_1^\infty \frac{\mathrm{d}r}{r^{m-1} f^*(r\varphi)} < \infty,$$

then δ_F is inadmissible.

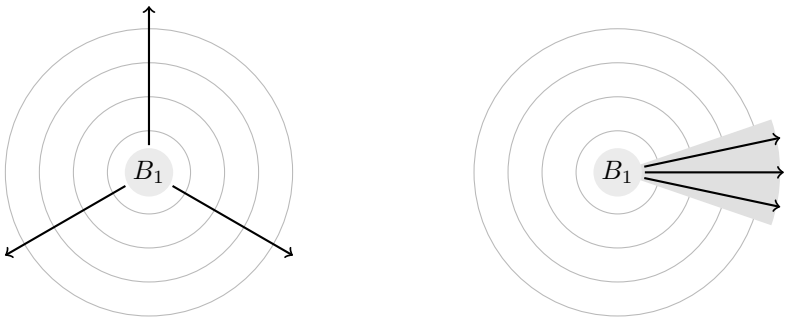
Brown's proofs are those of the corresponding geometric statements: (a) is the Dirichlet principle for the capacity of B_1; (d) uses the radial test functions $j_i(x) = c_i \int_{|x|}^{i} \mathrm{d}r/(r^{m-1}\bar{f}(r))$, whose energy tends to zero, exactly as in the sphere-area test of [12, Thm. 7.5] (the j_i are Lipschitz with compact support but not smooth; mollifying them gives functions of C_c^∞ whose energies still tend to zero, so they may be used in the definition of $\mathrm{cap}(B_1)$ above); (e) integrates a one-dimensional capacity bound along rays. (Brown's diffusion can explode in finite time if the drift grows faster than linearly; his correction [6] shows that when $\limsup_{|x|\to\infty} |\nabla \log f^*|/|x| < \infty$ there is no explosion and every result of [5] holds as stated. Every weight in this note has bounded drift by Lemma 2.3, so this case does not arise.) We say that the sphere test is *silent* on F when its integral converges, and that the ray test is silent when no set of directions of positive measure has uniformly bounded ray integrals. Brown's undecided zone is the set of F on which both are silent (Figure 2). Table 1 collects the dictionary.

Remark 2.2 (priority). Grigor'yan [12, p. 178] attributes the sphere-area test for Riemannian manifolds to Lyons–Sullivan [27] and to [14]; its classical ancestor for Riemann surfaces is the Ahlfors–Nevanlinna criterion of the 1930s. Brown's Theorem 6.4.3 is the same test on the weighted space W_F, with the same proof, carried out in 1971 in the statistical setting and so predating the general weighted-manifold statements; his Theorem 4.3.1 is the capacity characterization of parabolicity. In the sources we have read, neither literature cites the other.

2.3. **Explicit weights are marginals.** Brown's theorems concern Gaussian marginals $f^* = \varphi * F$. The weights we construct are explicit functions of the distance to a set, which are not Gaussian marginals as they stand. The following lemma removes the distinction.

TABLE 1. Brown's admissibility theory read as potential theory of $W_F = (\mathbb{R}^m, \delta, f^*\,\mathrm{d}x)$. Theorem numbers refer to [5] and [12].

statistics	weighted potential theory
generalized Bayes estimator $\delta_F = x + \nabla \log f^*$	weighted space W_F with measure $f^*\,\mathrm{d}x$
Brown's diffusion $\{Z_t\}$	weighted Brownian motion, generator $\Delta + \nabla \log f^* \cdot \nabla$
Brown's energy infimum (Thm. 4.3.1)	capacity of the unit ball (Grigor'yan, Thm. 5.1)
δ_F admissible, bounded drift (Thm. 5.1.1)	W_F parabolic
spherical F (Thm. 6.3.1)	model-manifold test $\int^\infty \mathrm{d}r/S(r)$, $S(r) = \omega_{m-1} r^{m-1} f^*(r)$
sphere test (Thm. 6.4.3)	sphere-area parabolicity test (Grigor'yan, Thm. 7.5)
—	ball-volume test $\int^\infty r\,\mathrm{d}r/V(r) = \infty$ (Thm. 7.3); an equivalence under D+P (Thm. 11.1)
—	flux test along an arbitrary exhaustion (Thm. 7.6); Lemma 5.1 below
MLE admissible iff $m \le 2$ (Stein [33])	Brownian motion recurrent iff $m \le 2$



(a) sphere test: shells in series $\sum \Delta r/(r^{m-1}\bar{f}) = \infty \Rightarrow$ recurrent

(b) ray test: a cheap cone of rays $\int \mathrm{d}r/(r^{m-1}f^*(r\varphi)) < \infty \Rightarrow$ transient

FIGURE 2. Brown's two radial tests as estimates of the resistance from B_1 to infinity. The sphere test adds the resistances of spherical shells in series, a lower bound whose divergence forces recurrence; the ray test finds a cone of rays of small total resistance, an upper bound whose finiteness forces transience. A weight whose spheres conduct well on average while every ray is expensive leaves both silent: Brown's undecided zone, the subject of this note.

Lemma 2.3 (comparison)**.** *Let $w > 0$ on $\mathbb{R}^m$ be locally integrable with $\log w$ Lipschitz with constant L, and let $f^* = \varphi * w$, the marginal of the prior $F = w\,\mathrm{d}\theta$. Then*

(i) *$c_L\, w \le f^* \le C_L\, w$ with $c_L = E\, e^{-L|Z|} > 0$ and $C_L = E\, e^{L|Z|} < \infty$, $Z \sim N(0, I_m)$;*
(ii) *$|\nabla \log f^*| \le L$;*
(iii) *F is σ-finite and $f^* < \infty$, so Theorem 2.1 applies to δ_F;*
(iv) *the spaces $(\mathbb{R}^m, w\,\mathrm{d}x)$ and $(\mathbb{R}^m, f^*\,\mathrm{d}x)$ have comparable Dirichlet forms, hence the same parabolicity type; doubling constants, Poincaré constants, sphere means, ray integrals and ball volumes of the two weights are comparable, so Brown's tests are silent for w if and only if they are silent for f^*, and w is D+P if and only if f^* is.*

Proof. From $e^{-L|z|} \le w(x+z)/w(x) \le e^{L|z|}$, integration against φ gives (i); $\nabla f^* = \varphi * (w\nabla \log w)$ and $|\nabla \log w| \le L$ give (ii); (iii) is immediate, and (iv) follows from (i) applied to $\int |\nabla u|^2 w\,\mathrm{d}x$ and to the integrals that define the other quantities, together with Theorem 2.1(a). □

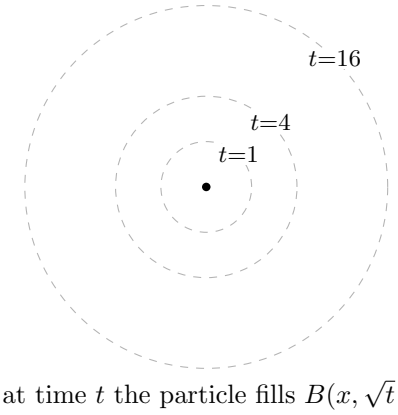
For $m = 3$ and $L = 2$ one has $C_L = 73.8$ and $c_L = 0.085$. Some of the weights below are marginals exactly: for $y, Z \in \mathbb{R}^2$, $E|y+Z|^4 = |y|^4 + 8|y|^2 + 8$, so the weight $(1+|y|^2)^2$ is, up to constants, the marginal of $\mathrm{dist}(\theta, \ell)^4\,\mathrm{d}\theta$ for a line $\ell \subset \mathbb{R}^3$.

2.4. **The volume criterion under doubling and Poincaré.** The following is the theorem of the geometric literature on which Theorem 3.1 rests.

Theorem 2.4 (Sturm; Grigor'yan; Saloff-Coste)**.** *Let w be a continuous positive weight on $\mathbb{R}^m$ that is D+P. Then $(\mathbb{R}^m, w\,\mathrm{d}x)$ is parabolic if and only if*

$$\int_1^\infty \frac{r\,\mathrm{d}r}{V(r)} = \infty. \tag{2}$$

Sources. Since w is continuous and positive, it is bounded above and below on every ball. Hence the form $\mathcal{E}$ on $C_c^\infty(\mathbb{R}^m)$ is closable in $L^2(\mu)$, and its closure is a regular, strongly local, irreducible Dirichlet form, with domain the functions $u \in L^2(\mu)$ whose distributional gradient lies in $L^2(\mu)$, with energy measure $|\nabla u|^2\,\mathrm{d}\mu$, and with the Euclidean distance as intrinsic metric. This is the setting of Sturm [34, 35, 36]; compare [36, Example 5 B]. Closed balls are compact, doubling is his condition (Ib), and (1), which passes from C^1 functions to the whole domain because the domain



return density $p_t(x,x) \asymp 1/V(x,\sqrt{t}\,)$

$$\int_1^\infty p_t(x,x)\ \mathrm{d}t \ \asymp\ \int_1^\infty \frac{r\ \mathrm{d}r}{V(r)}$$

diverges $\Rightarrow$ recurrent (under D+P)

FIGURE 3. The volume test (Theorem 2.4) from the heat kernel. Under doubling and a Poincaré inequality the return density is $p_t(x,x) \asymp 1/V(x,\sqrt{t}\,)$, so the expected time the particle spends near its start, $\int p_t(x,x)\ \mathrm{d}t \asymp \int r\ \mathrm{d}r/V(r)$, is infinite exactly when $(\mathbb{R}^m, w\,\mathrm{d}x)$ is recurrent.

is a closure, gives his Poincaré inequality (Ic). Under these conditions doubling and the Poincaré inequality are equivalent to the parabolic Harnack inequality [36, Thm. 3.5], the heat kernel has two-sided Gaussian bounds, in particular $p(t,x,x) \asymp 1/V(x,\sqrt{t})$ (Figure 3), and the Green function satisfies

$$g(x,y) \ \asymp\ \int_{|x-y|}^\infty \frac{r\,\mathrm{d}r}{\mu(B(x,r))}$$

[36, Cor. 4.11]; consequently the form is recurrent if and only if (2) holds [36, Cor. 4.12], [35, Cor. 2.9]. Recurrence is parabolicity in the sense of Section 2.1. Directly: if the integral in (2) converges, the upper bound for g gives $\int_{B_1} G1_{B_1}\,\mathrm{d}\mu < \infty$, where G is the Green operator, and then $\mu(B_1)^2 \le \big(\int_{B_1} G1_{B_1}\,\mathrm{d}\mu\big)\,\mathcal{E}(u)$ for every test function $u \ge 1$ on B_1 by [11, Lemma 1.5.3], so $\mathrm{cap}(B_1) > 0$. For smooth weights the same results are in [31] (Theorems 3.1 and 4.2, and the Green function criterion on p. 34) and [15], and for weighted manifolds in the survey [13]; for Riemannian manifolds see [12, Thm. 11.1]. The implication "(2) $\Rightarrow$ parabolic" needs neither doubling nor the Poincaré inequality: it follows from Corollary 2.5 below and the sphere test, Theorem 2.1(d) (compare [12, Thm. 7.3] for complete Riemannian manifolds). □

Corollary 2.5 (the sphere test dominates the volume test)**.** *For any positive weight, divergence of* $\int^\infty r\,\mathrm{d}r/V(r)$ *implies divergence of* $\int^\infty \mathrm{d}r/(r^{m-1}\bar f(r))$.

Proof. $V'(r) = \omega_{m-1} r^{m-1}\bar f(r)$. By the Cauchy–Schwarz inequality on $[R,2R]$, $R^2 = \big(\int_R^{2R}\mathrm{d}r\big)^2 \le \int_R^{2R} V'\,\mathrm{d}r \cdot \int_R^{2R} \mathrm{d}r/V' \le V(2R)\int_R^{2R}\mathrm{d}r/V'$, while $\int_R^{2R} r\,\mathrm{d}r/V \le 2R^2/V(R)$. Summing over $R = 2^j$, divergence of the volume integral forces $\sum_j 4^j/V(2^{j+1}) = \infty$ and hence divergence of $\int \mathrm{d}r/V'$. □

3. The reach of the radial tests

Theorem 3.1 (the sphere test is complete under D+P)**.** *Let w be a continuous positive weight on $\mathbb{R}^m$, $m \ge 2$, that is D+P. Then the following are equivalent:*

(i) $(\mathbb{R}^m, w\,\mathrm{d}x)$ *is parabolic;*
(ii) $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar f(r)) = \infty$ *(Brown's sphere test fires);*
(iii) $\int_1^\infty r\,\mathrm{d}r/V(r) = \infty$ *(the ball-volume test fires).*

Consequently, if f^* is D+P with bounded drift, δ_F is admissible if and only if Brown's sphere test fires. The ray test implies transience but is not necessary for it: the weight of Theorem 4.1 is D+P and transient with the ray test silent.

Proof. (iii)⇒(ii) is Corollary 2.5; (ii)⇒(i) is Brown's sphere test, Theorem 2.1(d); and (i)⇒(iii) is the converse direction of Theorem 2.4. The statement about δ_F is Theorem 2.1(b) with Lemma 2.3 if f^* is only comparable to a D+P weight. The last sentence is Theorem 4.1. □

Thus, within the D+P class, the geometric literature adds nothing on the admissible side, where Brown's 1971 sphere test is necessary and sufficient, and it can add exactly one kind of case on the inadmissible side: a transient weight along whose every ray the escape to infinity is expensive. Theorem 4.1 shows that this case occurs. The next two lemmas locate the undecided zone more precisely and explain why one spatial dimension is not enough to enter it.

Lemma 3.2 (bounded spherical A_2 ratio)**.** *Suppose $\sup_{r\ge1} Q(r) = C < \infty$. If $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar f(r)) = \infty$ then $(\mathbb{R}^m, w\,\mathrm{d}x)$ is parabolic; if the integral converges then the ray test fires. Hence Brown's two tests are exhaustive on weights with bounded spherical A_2 ratio, and every weight in the undecided zone has $Q(r)\to\infty$ along a sequence.*

Proof. The first case is Theorem 2.1(d). In the second, let $R(\varphi) = \int_1^\infty \mathrm{d}r/(r^{m-1}w(r\varphi))$. By Fubini its mean over the unit sphere is $\int_1^\infty A(r)\,\mathrm{d}r/r^{m-1} \le C\int_1^\infty \mathrm{d}r/(r^{m-1}\bar f(r)) < \infty$, so by Chebyshev's inequality the set $\{\varphi : R(\varphi) \le 2\,\mathrm{mean}\,R\}$ has at least half the surface measure, and R is bounded on it: the hypothesis of Theorem 2.1(e). □

Lemma 3.3 (one-dimensional D+P forces A_2)**.** *Let $v > 0$ be continuous on $\mathbb{R}$ with $\mu_1 = v\,\mathrm{d}t$ doubling, $\mu_1(2I) \le D\mu_1(I)$ for all intervals I, and satisfying $\int_I |u-u_I|^2\,\mathrm{d}\mu_1 \le C|I|^2\int_I |u'|^2\,\mathrm{d}\mu_1$ for all intervals. Then, with $\nu_1 = v^{-1}\,\mathrm{d}t$, $\mu_1([-r,r])\,\nu_1([-r,r]) \le 32\,CD^3r^2$ for every $r>0$. Consequently the product weight $w(x) = v(x_1)$ on $\mathbb{R}^m$, $m\ge3$, has bounded spherical A_2 ratio, and δ_F is inadmissible for every prior F whose marginal is comparable to w (Lemma 2.3; e.g. $\log v$ Lipschitz). That one-dimensional p-admissible measures are exactly the Muckenhoupt A_p weights is due to Björn, Buckley and Keith [4]; we record the case $p=2$ with the explicit constant used below.*

Proof. Take $I = [-2r, 2r]$ and the test function $u = 0$ on $[-2r,-r]$, $u = 1$ on $[r,2r]$, and $u(t) = \nu_1([-r,t])/\nu_1([-r,r])$ in between, which minimizes the energy across the middle segment: $\int_I |u'|^2\,\mathrm{d}\mu_1 = 1/\nu_1([-r,r])$. This u is Lipschitz and C^1 except at $\pm r$, where u' jumps; the Poincaré inequality, assumed for C^1 functions, extends to it by smoothing u' near $\pm r$, which changes both sides by an arbitrarily small amount. A function equal to 0 on a set of measure a and to 1 on a set of measure b has weighted variance at least $ab/(a+b) \ge \frac12\min(a,b)$. Since $[-r,r]$ is contained in the eightfold dilates of $[r,2r]$ and of $[-2r,-r]$, doubling gives $\mu_1([r,2r]) \ge \mu_1([-r,r])/D^3$ and likewise on the left, and the Poincaré inequality reads $\mu_1([-r,r])/(2D^3) \le C(4r)^2/\nu_1([-r,r])$. For the product weight, the sphere means are weighted means of v and $1/v$ over $[-r,r]$ with a kernel bounded above and bounded below on $[-r/\sqrt2, r/\sqrt2]$, so by doubling $\bar f(r) \asymp \mu_1([-r,r])/r$ and $A(r) \lesssim \nu_1([-r,r])/r$, whence Q is bounded. Doubling also gives reverse doubling, $\mu_1([-r,r]) \ge cr^\varepsilon$, so $\int^\infty \mathrm{d}r/(r^{m-1}\bar f) \lesssim \int^\infty r^{2-m-\varepsilon}\,\mathrm{d}r < \infty$ for $m\ge3$, and Lemma 3.2 gives inadmissibility. □

Both hypotheses are needed: $v(t) = (1+|t|)^2$ is doubling but fails the Poincaré inequality, because the valley at 0 is a bottleneck between two heavy sides, and $v(t) = (1+|t|)^{-1}$ satisfies the Poincaré inequality but is not doubling; $\mu_1\nu_1/r^2$ grows like r in the first case and like $\log r$ in the second (Figure 8b,c). The mechanism of Lemma 3.3 is that a valley in one dimension has co-dimension one and disconnects the interval. In $\mathbb{R}^m$, $m\ge3$, a valley along a curve has co-dimension two, does not disconnect, and the lemma fails:

Theorem 3.4 (co-dimension-two valleys)**.** *For $m \geq 3$ and $\alpha \geq 0$ the weight $w(x) = (1 + x_1^2 + \cdots + x_{m-1}^2)^{\alpha/2}$ on $\mathbb{R}^m$, a power of the distance to the x_m-axis, is D+P. For $\alpha > m-1$ its spherical A_2 ratio satisfies $Q(r) \asymp r^{\alpha-(m-1)} \to \infty$.*

Proof. Björn [3, Prop. 6], generalizing Chanillo and Wheeden, proves that for $x = (x', x'')$ with $x' \in \mathbb{R}^k$, $2 \leq k \leq n$, any $v \in A_p$, points $a_j \in \mathbb{R}^k$ and exponents $\gamma_0, \gamma_j \geq 0$, the weight $v(x)(1 + |x'|)^{\gamma_0} \prod_j \big(|x' - a_j|/(1 + |x' - a_j|)\big)^{\gamma_j}$ on $\mathbb{R}^n$ is p-admissible, that is, doubling and supporting the weak $(1,p)$-Poincaré inequality. Take $n = m$, $k = m-1$, $v \equiv 1$, $\gamma_0 = \alpha$, no a_j, and $p = 2$; since $(1+|x'|)^\alpha \asymp (1+|x'|^2)^{\alpha/2}$, the weight is doubling and supports the weak $(1,2)$-Poincaré inequality in the sense of [17, (5), p. 7] for locally Lipschitz functions, in particular for each pair $(u, |\nabla u|)$ with u of class C^1. Under doubling this self-improves to the strong $(2,2)$ form (1): [17, Thm. 5.1] raises the exponent on the left to any $h < sp/(s-p)$ when $p < s$, with the energy on a dilated ball, and [17, Cor. 9.8] (their Theorem 9.7 for balls) removes the dilation. Here $p = 2$; the exponent s in the lower mass bound exists for every doubling measure and may be taken larger than 2 [17, Lemma 14.6]; and $h = 2 < 2s/(s-2)$, so the truncation property, needed only at the endpoint exponent, is not involved. The constant in (1) obtained in this way depends only on the doubling constant and on the constant of the weak inequality [17, pp. 23 and 44]; we use this uniformity in Lemma 4.3. For the asymptotics parameterize the sphere $\{|x| = r\}$ by $|x'| = r\sin\theta$; then $\bar{f}(r) \asymp r^\alpha$, and for $\alpha > m-1$ the integral of $1/w$ concentrates on the polar caps $\theta \lesssim 1/r$, giving $A(r) \asymp r^{-(m-1)}$. □

So Lemma 3.3 is a one-dimensional statement, and weights in the undecided zone that are D+P must have valleys of co-dimension at least two. The ray test, however, is silent only if *every* direction (up to a null set) meets cheap regions at infinitely many scales, which a single line does not achieve. The construction of the next section makes the valleys sweep every direction without ever forming a surface (Figure 4).

4. A doubling–Poincaré prior that neither test decides

4.1. **The construction.** Work in $\mathbb{R}^3$ with spherical coordinates (r, θ, ϕ), $\theta \in [0, \pi]$ the polar angle. For $k \geq 4$ put

$$N_k = 2^{k+2}, \qquad L_k = 2^{\lceil k/2 \rceil}, \qquad M_k = N_k/L_k, \qquad s_k = 2^{k/2}.$$

Shell k is $\{2^k \leq |x| < 2^{k+1}\}$. For $0 \leq i < L_k$ and $0 \leq j < M_k$ let $n = jL_k + i$ and let $\Gamma_{k,i,j}$ be the latitude circle

$$\Gamma_{k,i,j} = \Big\{ |x| = 2^k + i\,\frac{2^k}{L_k}, \quad \theta = \theta_n := \frac{\pi(n + \frac{1}{2})}{N_k} \Big\}.$$

Since $(i,j) \mapsto n$ is a bijection onto $\{0, \ldots, N_k - 1\}$, the N_k circles of shell k occupy *all* the polar angles $\pi(n+\frac{1}{2})/N_k$, and every direction is within $\pi/(2N_k)$ in angle of one of them. In the meridian half-plane $(s, z) = (r\sin\theta, r\cos\theta)$ the circles are points forming a sheared rectangular lattice: L_k layers at radial spacing $2^k/L_k \in \{s_k, s_k/\sqrt{2}\}$, each containing M_k circles at angular spacing $\pi L_k/N_k$, that is, at physical spacing between $0.78\,s_k$ and $2.23\,s_k$; consecutive layers are offset by one angular step π/N_k (Figure 7a). Let $\Gamma = \bigcup_{k,i,j} \Gamma_{k,i,j}$ and define, for $\alpha \geq 0$,

$$w(x) = \Big(\frac{1 + d(x)^2}{1 + |x| + d(x)^2} \Big)^{\alpha/2}, \qquad d(x) = \operatorname{dist}(x, \Gamma). \tag{3}$$

Write $s(r)^2 = 1 + r$. Near a circle at radius r, $w \asymp (1 + d^2)^{\alpha/2}/s^\alpha$ for $d \leq s/2$: the cylinder weight of Theorem 3.4 normalized by the local spacing. For $d \geq s$, w is bounded below by $2^{-\alpha/2}$. The set where w is small is therefore a union of thin tubes around circles that are $\asymp s_k$ apart; no surface of small weight exists at any scale. Two features of the construction are forced. The spacing $s_k \asymp 2^{k/2}$ comes from counting: $N_k \asymp 2^k$ circles of length $\asymp 2^k$ must be spread through a shell of volume $\asymp 8^k$ at mutual distance s, so $s^2 \asymp 8^k/4^k$. The exponent $\alpha \geq 4$ is the threshold at which a ray that misses a circle by a bounded distance still pays a constant per shell, see (4).



(a) a wall, co-dimension one
$\lambda_1\rho^2 \to 0$: Poincaré fails

(b) scattered curves, co-dimension two
$\lambda_1\rho^2 \gtrsim 1$: Poincaré holds

FIGURE 4. Why the circles must be scattered. On a ball, a thin low-weight surface of co-dimension one (a wall) splits it into two rooms joined only across the wall, and the Poincaré constant $1/(\lambda_1\rho^2)$ grows without bound as the ball grows (Proposition 5.2). Thin low-weight tubes around curves (co-dimension two) do not disconnect the ball, and the Poincaré inequality survives with constants uniform in the scale (Theorem 4.1).

The function $\log w$ is Lipschitz: $\nabla \log w = \frac{\alpha}{2}\big[2d\nabla d/(1+d^2) - (\nabla|x| + 2d\nabla d)/(1+|x|+d^2)\big]$ with $|\nabla d| = |\nabla|x|| = 1$ almost everywhere, so, writing $A = 1+d^2$ and $B = 1+|x|+d^2$,

$$|\nabla \log w| \le \frac{\alpha}{2}\big[2d(A^{-1} - B^{-1}) + B^{-1}\big] = \frac{\alpha}{2}\,\frac{2d|x| + 1 + d^2}{(1+d^2)(1+|x|+d^2)} \le \frac{\alpha}{2},$$

as $2d \le 1 + d^2$. By Lemma 2.3, $f^* = \varphi * w$ is a smooth Gaussian marginal comparable to w with drift bounded by $\alpha/2$.

Theorem 4.1 (the lattice prior)**.** *Let w be the weight (3) on $\mathbb{R}^3$, $\alpha \ge 4$, and $F = w\,\mathrm{d}\theta$. Then:*

(a) *$w\,\mathrm{d}x$ is doubling and supports the scale-invariant Poincaré inequality (1) on all balls (this part holds for every $\alpha \ge 0$);*
(b) *Brown's sphere test is silent, $\bar f(r) \ge c > 0$ for $r \ge 16$; Brown's ray test is silent, and in fact $\int_{16}^{\infty} \mathrm{d}r/(r^2 w(r\varphi)) = \infty$ for* every *direction φ;*
(c) *$V(r) \asymp r^3$.*

Consequently $(\mathbb{R}^3, f^*\,\mathrm{d}x)$ is transient and δ_F is inadmissible, although neither of Brown's radial tests applies.

Proof of (b), (c) and of the consequence. Fix a direction φ and a shell $k \ge 5$. Some θ_n is within $\delta \le \pi/(2N_k)$ of the polar angle of φ; let $r_0 < 2^{k+1}$ be the radius of the corresponding circle of shell k. The point of that circle in the meridian half-plane of φ is at distance $\big((r-r_0)^2 + 2rr_0(1-\cos\delta)\big)^{1/2}$ from $r\varphi$, and $2rr_0(1-\cos\delta) \le rr_0\delta^2 \le \pi^2/16$ for $r \le 2^{k+1}$. So for $r \in [r_0 - 1, r_0 + 1]$ one has $d^2 \le 1 + \pi^2/16 < 1.62$ and hence $w \le (2.62/r_0)^{\alpha/2}$, so

$$\int_{r_0-1}^{r_0+1} \frac{\mathrm{d}r}{r^2 w(r\varphi)} \;\ge\; \frac{2\,(r_0/2.62)^{\alpha/2}}{(r_0+1)^2} \;\ge\; 2.62^{-\alpha/2}\, r_0^{\alpha/2-2}, \tag{4}$$

using $2(r_0/(r_0+1))^2 \ge 1$ for $r_0 \ge 16$. The coefficient $2.62^{-\alpha/2}$ depends on α: the per-shell toll is a positive constant (at least 0.14) for $\alpha = 4$ and grows like $2^{k(\alpha/2-2)}$ for $\alpha > 4$. Summing over k the ray integral diverges, for every φ. For the sphere test, on each sphere $\{|x| = r\}$ with r in shell k the set of points at distance $\ge s_k/4$ from Γ has a surface-measure fraction bounded below by a constant c_0: the layers of circles are $\ge s_k/2$ apart radially (including those of the neighbouring shells), so the tubes of radius $s_k/4$ of at most one layer meet the sphere in positive area, and the circles of that layer, at angular spacing $\pi L_k/N_k$, meet it in bands of angular width $\le s_k/(2r)$, which cover an angular fraction $\le s_k N_k/(2\pi r L_k) \le 2/\pi$; hence $c_0 \ge 1 - 2/\pi - o(1)$. There, since

w is increasing in d and $r < 2s_k^2$, $w \geq \big((s_k/4)^2/(1+r+s_k^2/16)\big)^{\alpha/2} \geq 34^{-\alpha/2}$. So $\bar f(r) \geq c_0 34^{-\alpha/2}$ and the sphere integral $\int \mathrm{d}r/(r^2\bar f)$ converges. For (c), $w \leq 1$ gives $V(r) \leq \frac{4}{3}\pi r^3$, and the same fraction argument applied to shells gives $V(r) \geq cr^3$. Finally, by (a) and (c), $\int^\infty r\,\mathrm{d}r/V(r) < \infty$, so $(\mathbb{R}^3, w\,\mathrm{d}x)$ is transient by Theorem 2.4; by Lemma 2.3(iv) so is $(\mathbb{R}^3, f^*\,\mathrm{d}x)$, and δ_F is inadmissible by Theorem 2.1(b). □

4.2. **Proof of (a): doubling and Poincaré.** The proof assembles the Poincaré inequality on a large ball from the inequality on cells containing at most one circle (Figure 5), where the weight is the cylinder weight of Theorem 3.4 up to constants. We first record the assembly lemma and then the inequality for one circle. The assembly lemma is a variant of the discretization technique of Kanai [20] and of Coulhon and Saloff-Coste [8, Prop. 6.10, Thm. 7.1], in the form given by Grigor'yan and Saloff-Coste [16, Thm. 3.7], who glue Poincaré inequalities over a covering by means of a discrete Poincaré inequality on the graph of the covering. We include a proof because we need cells of variable size and obtain the discrete inequality from a comparison measure.

Lemma 4.2 (gluing)**.** *Let μ and ν be measures on $\mathbb{R}^n$ with continuous nonnegative densities, and let $B = B(x_0,\rho)$. Suppose that ν satisfies (1) on the ball $2B$, and that the closed ball $2\bar B$ is covered by finitely many closed cubes Q_j with bounded overlap and side lengths s_j with $\operatorname{diam} Q_j = \sqrt{n}\,s_j \leq \rho$; write Q_j^* for the concentric 1.01-fold enlargement of Q_j and $j' \sim j$ if $Q_{j'}^* \cap Q_j^* \neq \emptyset$. Suppose that $s_j \asymp s_{j'}$ whenever $j' \sim j$, and that for each j there is a ball $B_j \supset \bigcup_{j'\sim j} Q_{j'}^*$ of radius $\leq Cs_j$ with $B_j \subset 4B$, the balls B_j having bounded overlap, such that*

(G1) *$\mu(Q_j) \asymp \nu(Q_j)$ for all j, and*

(G2) *μ satisfies (1) on B_j and is doubling at every ball centred in B_j of radius at most that of B_j, all with constants independent of j and of B. Then $\int_B |u-u_B|^2\,\mathrm{d}\mu \leq C'\rho^2 \int_{4B}|\nabla u|^2\,\mathrm{d}\mu$ for every $u \in C^1(4\bar B)$, with C' depending only on n and the constants above. (The inclusions $B_j \subset 4B$ are a hypothesis on every cube of the family: they do not follow from the covering and $\operatorname{diam} Q_j \leq \rho$, and are verified wherever the lemma is applied.)*

Proof. Let $\bar u_j$ be the μ-mean of u on Q_j. Let $\{\psi_j\}$ be smooth functions with values in $[0,1]$, ψ_j vanishing outside Q_j^*, with $\sum_j \psi_j = 1$ on a neighbourhood of $\bigcup_j Q_j$ and $|\nabla\psi_{j'}| \lesssim 1/s_j$ on Q_j for every $j' \sim j$. Such functions exist: by a volume count, the hypotheses (bounded overlap of the cubes, comparable sides of neighbours) give bounded overlap of the enlarged cubes and a bound on the number of neighbours of each cube. Both the identity $\sum_j \psi_j = 1$ and the gradient bound are asked on the cubes Q_j only: a family that sums to one on all of the enlarged cubes cannot satisfy such a bound there uniformly. Let $U = \sum_j \bar u_j \psi_j$, a C^1 function on $2\bar B$. For any constant m, $\int_B |u-m|^2\,\mathrm{d}\mu \leq 2\int_B |u-U|^2\,\mathrm{d}\mu + 2\int_B |U-m|^2\,\mathrm{d}\mu$. On Q_j, $|u-U| \leq \sum_{j'\sim j}\psi_{j'}|u-\bar u_{j'}|$. For $j' \sim j$ the ball $B_{j'}$ contains Q_j and $Q_{j'}$, and $\mu(Q_j) \leq \mu(B_{j'}) \lesssim \mu(Q_{j'})$ by (G2); comparing $\bar u_{j'}$ with the mean of u over $B_{j'}$ and using (1) on $B_{j'}$, $\int_{Q_j}|u-\bar u_{j'}|^2\,\mathrm{d}\mu \lesssim s_j^2 \int_{B_{j'}}|\nabla u|^2\,\mathrm{d}\mu$; summing with bounded overlap, the first term is $\lesssim \rho^2\int_{4B}|\nabla u|^2\,\mathrm{d}\mu$. For the second term, on Q_j one has $|U-m|^2 \leq 2|\bar u_j - m|^2 + 2|U-\bar u_j|^2$ and $|U-\bar u_j| \leq \sum_{j'\sim j}|\bar u_{j'} - \bar u_j|$, so by (G1)

$$\int_B |U-m|^2\,\mathrm{d}\mu \;\lesssim\; \sum_j \nu(Q_j)|\bar u_j - m|^2 + \sum_j \nu(Q_j) \sum_{j'\sim j} |\bar u_{j'} - \bar u_j|^2.$$

The sums run over the cubes meeting B, which lie in $2B$ since $\operatorname{diam} Q_j \leq \rho$. Choosing m to be the ν-mean of U on $2B$ and using $\nu(Q_j)|\bar u_j - m|^2 \leq 2\int_{Q_j}|U-m|^2\,\mathrm{d}\nu + 2\nu(Q_j)\sup_{Q_j}|U-\bar u_j|^2$ together with the Poincaré inequality for ν on $2B$ and $|\nabla U|^2 \lesssim s_j^{-2}\sum_{j'\sim j}|\bar u_{j'} - \bar u_j|^2$ on Q_j, everything is bounded by $\rho^2 \sum_j \nu(Q_j) s_j^{-2} \sum_{j'\sim j}|\bar u_{j'} - \bar u_j|^2$, now summed over the cubes meeting $2B$ (the factor ρ^2 absorbs the second sum since $s_j \leq \rho$). Finally, for $j' \sim j$ both cubes lie in B_j, and

by (G2) $|\bar u_{j'}-\bar u_j|^2 \lesssim \mu(Q_j)^{-1}\int_{B_j}|u-u_{B_j}|^2\,\mathrm{d}\mu \lesssim s_j^2\mu(Q_j)^{-1}\int_{B_j}|\nabla u|^2\,\mathrm{d}\mu$ (local doubling makes $\mu(B_j)\lesssim\mu(Q_j)$). With $\nu(Q_j)\asymp\mu(Q_j)$ the factors $s_j^{\pm2}$ and $\mu(Q_j)^{\pm1}$ cancel, and bounded overlap of the B_j gives the claim. □

The passage from energy over $4B$ to (1) is not a statement about one ball. If μ is doubling and satisfies the inequality with energy over $4B$ at *every* ball, with constants independent of the ball, then μ satisfies (1) at every ball, with a constant depending only on those two constants [17, Thm. 5.1 and Cor. 9.8, with the statements on the constants on pp. 23 and 44]; see also [36, Thm. 2.4]. We refer to this as the *self-improvement*. In both applications below, doubling of μ at all balls is verified directly.

Lemma 4.3 (one circle)**.** *Let $C_R\subset\mathbb{R}^3$ be a circle of radius $R\ge0$ and $w_R(x)=(1+\operatorname{dist}(x,C_R)^2)^{\alpha/2}$, $\alpha\ge0$. Then $w_R\,\mathrm{d}x$ is D+P with constants that depend only on α; in particular they are independent of R.*

Proof. All constants depend only on α. Write $\mu_R=w_R\,\mathrm{d}x$, $B=B(x_0,\rho)$ and $\tau=\operatorname{dist}(x_0,C_R)$.

Step 1: doubling, and the case $\tau>2\rho$. If $\tau>2\rho$ then $\operatorname{dist}(\cdot,C_R)\in[\tau/2,3\tau/2]$ on B and $\le2\tau$ on $2B$, so w_R varies on B by a factor $\le9^{\alpha/2}$ and $\sup_{2B}w_R\le16^{\alpha/2}\inf_B w_R$: μ_R is comparable to Lebesgue measure on B, and (1) and doubling hold at B. If $\tau\le2\rho$ then $\operatorname{dist}(\cdot,C_R)<4\rho$ on $2B$, so $\mu_R(2B)\le8|B|(1+16\rho^2)^{\alpha/2}$; and the ball of radius $\rho/4$ about the point at distance $\rho/2$ from x_0 in the direction perpendicular to the plane of C_R, away from that plane, lies in B at distance $\ge\rho/4$ from the plane, hence from C_R, so $\mu_R(B)\ge\frac{1}{64}|B|(1+\rho^2/16)^{\alpha/2}$ and $\mu_R(2B)\le512\cdot16^{\alpha}\mu_R(B)$. Thus μ_R is doubling at every ball.

Step 2: scales $\rho\le R/6$, the case $\tau\le2\rho$. Every $y\in2B$ has $\operatorname{dist}(y,C_R)<4\rho\le2R/3$. In cylindrical coordinates (r,ϕ,z) about the axis of C_R, with $C_R=\{r=R,\ z=0\}$ and $\phi=0$ at x_0, put $t=R\phi$ and $\eta=(R-r,z)$. The map $\Psi\colon x\mapsto(t,\eta)$ is a diffeomorphism of $\{r>0,\ |\phi|<\pi\}$ onto $\{|t|<\pi R,\ \eta_1<R\}$ with $|\eta|=\operatorname{dist}(\cdot,C_R)$, and its differential has singular values $R/r,1,1$. So Ψ carries w_R exactly to the cylinder weight $(1+|\eta|^2)^{\alpha/2}$ of Theorem 3.4, and the measure μ_R to r/R times the cylinder measure $\mu_{\mathrm{cyl}}=(1+|\eta|^2)^{\alpha/2}\,\mathrm{d}t\,\mathrm{d}\eta$, which satisfies (1) with a constant P_{cyl} by Theorem 3.4. On B one has $r\ge R-3\rho\ge R/2$, so Ψ is 2-Lipschitz on the convex set B and $\Psi(B)$ lies in the ball $\widehat B$ of radius 2ρ about $\Psi(x_0)$. On $\widehat B$, $|\eta_1|<4\rho\le2R/3$ and $|t|<2\rho\le R/3$, so Ψ^{-1} is defined on $\widehat B$ with no periodic identification, $r/R\in[\frac13,\frac53]$ there, Ψ^{-1} is $\frac53$-Lipschitz on $\widehat B$, and $\Psi^{-1}(\widehat B)\subset\frac{10}{3}B\subset4B$. For $u\in C^1(4\bar B)$ put $v=u\circ\Psi^{-1}$ on the closure of $\widehat B$ and let a be the μ_{cyl}-mean of v on $\widehat B$. Since $r/R\le\frac32$ on B and $\max(r/R,R/r)\le3$ on $\Psi^{-1}(\widehat B)$,

$$\int_B|u-a|^2\,\mathrm{d}\mu_R\le\tfrac32\int_{\widehat B}|v-a|^2\,\mathrm{d}\mu_{\mathrm{cyl}}\le6P_{\mathrm{cyl}}\rho^2\int_{\widehat B}|\nabla v|^2\,\mathrm{d}\mu_{\mathrm{cyl}}\le18P_{\mathrm{cyl}}\rho^2\int_{4B}|\nabla u|^2\,\mathrm{d}\mu_R.$$

The same inequality, with energy over $4B'$, holds at every ball B' with $4B'\subset B$: either its centre is at distance more than twice its radius from C_R and Step 1 applies, or B' is a ball of the kind just treated. As μ_R is doubling at every ball, the pair $(u,|\nabla u|)$ satisfies the weak $(1,2)$-Poincaré inequality in B in the sense of [17, (5), p. 7], and [17, Thm. 9.7], applied with $\Omega=B$ (a ball is a John domain [17, Cor. 9.5]), gives (1) on B. So (1) holds at every ball of radius $\rho\le R/6$.

Step 3: scales $\rho\ge R/6$. Let c be the centre of C_R and $p_c(x)=(1+|x-c|^2)^{\alpha/2}$. It is D+P: doubling and the weak $(1,2)$-Poincaré inequality are [3, Prop. 6] with $k=n=3$, and (1) follows by the self-improvement, as in the proof of Theorem 3.4; alternatively, [16, Prop. 4.9 with Thm. 2.7] gives both directly. If $R\le16$, then $||x-c|-R|\le\operatorname{dist}(x,C_R)\le|x-c|+R$ gives $1+\operatorname{dist}(x,C_R)^2\asymp1+|x-c|^2$ with absolute constants, so $w_R\asymp p_c$ and $w_R\,\mathrm{d}x$ is D+P; so assume $R>16$ (only $R>0$ is used below). Apply Lemma 4.2 with $\nu=p_c\,\mathrm{d}x$ and the cubes of the grid of side $R/16$ that meet $2\bar B$; they have diameter $\sqrt3R/16<\rho$ and no point lies in more than 8 of them. Let B_j be the ball of radius $0.163R$ about the centre of Q_j: it contains the 1.01-fold enlargements of Q_j and of its 26 neighbours,

which lie in a concentric cube of side $3.01R/16$. Since $0.163R < R/6$, (G2) holds by Steps 1 and 2. The centre of Q_j is within $2\rho+\sqrt{3}R/32$ of x_0, so $B_j \subset B(x_0, 2\rho+0.22R) \subset 4B$ because $R \le 6\rho$; and no point lies in more than $6^3 = 216$ of the balls B_j. For (G1) we show $\mu_R(B') \asymp \nu(B')$ for every ball $B' = B(x,\rho')$ with $\rho' \ge R/16$; since a cube of side $R/16$ lies between the balls of radii $R/32$ and $R/16$ about its centre and both measures are doubling, this gives (G1). Write $D = |x-c|$, and recall that $R \le 16\rho'$. Both measures are $\lesssim \rho'^3 \max(1,\rho',D)^\alpha$ since $\operatorname{dist}(y,C_R) \le |y-c| + R \le D + 17\rho'$ on B'. For the lower bound, if $D \ge 34\rho'$ then $|y-c| \ge D-\rho' \ge D/2$ and $\operatorname{dist}(y,C_R) \ge D-\rho'-R \ge D/2$ on B'. If $D \le 34\rho'$, let x' be the point at distance $\rho'/2$ from x in the direction perpendicular to the plane of C_R and away from that plane. The ball $B(x',\rho'/4)$ lies in B' and at distance $\ge \rho'/4$ from the plane, hence from C_R and from c; so $w_R, p_c \ge (1+\rho'^2/16)^{\alpha/2} \gtrsim \max(1,\rho',D)^\alpha$ on a ball of volume $|B'|/64$. This is the lower bound in both cases.

Step 4: conclusion. By Steps 1 to 3, μ_R is doubling at every ball and satisfies the Poincaré inequality with energy over $4B$ at every ball B: for $\rho \le R/6$ even (1), and for $\rho \ge R/6$ by Lemma 4.2. The constants so far depend only on α. The self-improvement therefore gives (1) at every ball, with a constant that depends only on the doubling constant and on the constant of the weak inequality, hence only on α and not on R. □

Remark 4.4. For $R > 0$ and $\alpha > 0$ Lemma 4.3 can also be derived from the literature. The complement of a circle in $\mathbb{R}^3$ is a uniform domain [38, Remark 5.3.1], with a constant independent of the radius, so $\operatorname{dist}(\cdot,C_R)^\alpha\,\mathrm{d}x$ is doubling and supports a 1-Poincaré inequality with constants depending only on α by a theorem of Keith [23, Thm. 4], whose example is this domain. The quotient $(1+d^2)^{\alpha/2}/d^\alpha$, $d = \operatorname{dist}(\cdot,C_R)$, is an A_1 weight with respect to that measure, with constant $\le 2\cdot 45^\alpha$, so [3, Thm. 4] and the self-improvement apply. We keep the direct proof, which rests on [3] and [17] alone.

Proof of Theorem 4.1(a). Write $\sigma(x)^2 = 1+|x|$, so $\sigma \asymp s_k$ on shell k (indeed $\sigma \in [s_k, 1.44\,s_k]$ there for $k \ge 4$). Recall the separation of the circles. The distance between two coaxial circles equals the distance between their generating points in the meridian half-plane, and likewise the distance from a point to a circle about the axis is the distance between their meridian points; within shell m the layers are $\ge s_m/\sqrt{2}$ apart radially and the circles of one layer $\ge 0.78\,s_m$ apart, and the innermost layer of shell $m+1$ lies $\ge s_m/\sqrt{2}$ beyond the outermost layer of shell m. Hence two distinct circles lying in shells $\ge k$ are at distance $\ge 0.7\,s_k$, while every circle of a shell $< k$ lies in $\{|x| \le 2^k - s_k/2\}$ (the radial gap $2^{k-1}/L_{k-1}$ below shell k can be as small as $s_k/2$). On $\{|x| \le 32\}$ one has $33^{-\alpha/2} \le w \le 1$, so μ is comparable to Lebesgue measure on every ball of radius ≤ 8 centred in $\{|x| < 16\}$, and cells centred there satisfy (G1) and (G2) below trivially; balls of radius > 8 centred in $\{|x| < 16\}$ are covered by the large-ball argument, which uses only $\sigma(x_0) < 10\rho$. In the rest of the proof x_0 lies in a shell $k \ge 4$. We first fix the local form of w and the cell mass, then treat small and large balls.

Local comparison. Let $B = B(x_0,\rho)$ with $\rho \le \sigma_0/10$ in shell k, where $\sigma_0 = \sigma(x_0)$; note $\rho \le 0.15\,s_k$. Then $1+|x|$ changes on B by $\le 2\rho \le \sigma_0/5$ against $1+|x_0| \ge 2^k$, so $\sigma \asymp \sigma_0$ and $1+|x| \ge \frac12\sigma_0^2$ there. Let C be a circle nearest to x_0, $\tau = \operatorname{dist}(x_0,C)$, and $d_C = \operatorname{dist}(\cdot,C)$. If $\tau \le 2\rho$, then C lies in shell k or $k+1$: the circles of shells $< k$ are at distance $\ge s_k/2 > 2\rho$ from x_0, and at distance $\ge s_k/2 - \rho \ge 0.35\,s_k$ from B; those of shells $> k+1$ are farther still. Every other circle C' of a shell $\ge k$ has $\operatorname{dist}(x_0,C') \ge 0.7\,s_k - \tau$, hence $\operatorname{dist}(y,C') \ge 0.7\,s_k - 3\rho \ge 0.25\,s_k$ for $y \in B$. So C is the only circle within ρ of B; on B the distance $d = \operatorname{dist}(\cdot,\Gamma)$ equals d_C wherever $d_C \le 0.25\,s_k$, and at the remaining points of B (there are none unless $3\rho > 0.25\,s_k$) one has $0.25\,s_k \le d \le d_C < 3\rho \le 0.45\,s_k$. If $\tau > 2\rho$, then on B, $d \ge \tau - \rho$ and $d_C \le \tau + \rho \le 3(\tau-\rho)$, so $d \le d_C \le 3d$. In either case $(1+d^2)^{\alpha/2} \asymp (1+d_C^2)^{\alpha/2}$ on all of B. In the case $\tau \le 2\rho$ moreover $d \le 3\rho \le 0.3\,\sigma_0$ on B, so $1+|x|+d^2 \asymp \sigma_0^2$ (both $\ge \frac12\sigma_0^2$ and $\le 2\sigma_0^2$), whence

$$w \;\asymp\; \sigma_0^{-\alpha}\,(1+d_C^2)^{\alpha/2} \qquad \text{on } B \text{ whenever } \operatorname{dist}(x_0,\Gamma) \le 2\rho, \tag{5}$$

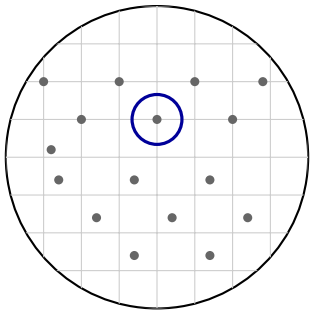
a *bounded-factor* comparison with constants depending only on α. (The second factor $(1+|x|+d^2)^{-\alpha/2}$ varies on B by a bounded factor, not by $1+O(1/s)$, because d^2 as well as $|x|$ varies across B.) In the case $\tau > 2\rho$ we use that $w = (1-|x|/(1+|x|+d^2))^{\alpha/2}$ is increasing in d and decreasing in $|x|$. For $y, z \in B$ one has $d(y) \le \tau+\rho \le 3(\tau-\rho) \le 3d(z)$, and for $y \in 2B$, $z \in B$, $d(y) \le \tau+2\rho \le 4(\tau-\rho) \le 4d(z)$; moreover $|z| \le |y|+3\rho$ and $\rho \le d(z) \le 1+d(z)^2$. Hence the ratio $(1+d^2)/(1+|x|+d^2)$ varies on B by a factor ≤ 9, and its supremum over $2B$ is at most 16 times its infimum over B; so w varies on B by a factor $\le 9^{\alpha/2}$ and $\sup_{2B} w \le 16^{\alpha/2} \inf_B w$.

Cell mass. Let Q be a closed cube of side ℓ, or a ball of radius ℓ, with $\ell \in [s_k/32,\, s_k]$ and centre x_Q in shell k. Let $B_0 \subset Q$ be the largest ball about x_Q contained in Q; its radius ρ_0 is $\ell/2$ or ℓ, and $|B_0| \ge \frac{1}{2}|Q|$. Put $r = s_k/64$, so that $r \le \rho_0$, and let e be the unit vector of the axis. We claim that for every $y \in B_0$ one of the three points y, $y+\frac{r}{2}e$, $y-\frac{r}{2}e$ lies at distance $\ge r/2$ from Γ. If $d(y) \ge r$, this point is y. Otherwise some circle C has a point within r of y. A second circle with a point within r of y would come within $2r = s_k/32$ of C. But both circles would lie in shells $\ge k-1$, because they come within $2s_k$ of x_Q and $2^k - 2s_k \ge 2^{k-1}$, and two distinct circles of shells $\ge k-1$ are at distance $\ge s_k/2$ (the separation statements above with $m = k-1$, as $s_{k-1}/\sqrt{2} = s_k/2$). Hence every circle other than C stays at distance $\ge r$ from y, and at distance $\ge r/2$ from $y \pm \frac{r}{2}e$. The circle C lies in a plane $\{x \cdot e = h\}$; with the sign chosen so that $y \pm \frac{r}{2}e$ is at distance $\ge r/2$ from this plane, the claim follows. Consequently the ball $B(x_Q, \rho_0 - r/2)$ is covered by G, $G+\frac{r}{2}e$ and $G-\frac{r}{2}e$, where $G = \{x \in B_0 :\ d(x) \ge r/2\}$, and so $|G| \ge \frac{1}{3}|B(x_Q, \rho_0/2)| = \frac{1}{24}|B_0| \ge \frac{1}{48}|Q|$. On G, $d \ge s_k/128$ gives, since w is increasing in d and $1+|x| \le 3s_k^2$ on Q, the bound $w \ge \big(\varepsilon^2/(4+\varepsilon^2)\big)^{\alpha/2}$ with $\varepsilon = 1/128$, that is, $w \ge 65537^{-\alpha/2} =: c_\alpha$. With $w \le 1$,

$$\tfrac{1}{48}\, c_\alpha\, |Q| \;\le\; \mu(Q) \;\le\; |Q| \qquad \text{for all such } Q. \tag{6}$$

Small balls ($\rho \le \sigma_0/10$). If $\tau > 2\rho$, then by the local comparison w varies on B by a bounded factor and $\sup_{2B} w \le 16^{\alpha/2} \inf_B w$: μ is comparable to Lebesgue measure on B, and (1) and doubling hold at B. Otherwise $\tau \le 2\rho$, the nearest circle C is the only one within ρ of B (the case $\tau \le 2\rho$ above), and its Euclidean radius R_C may be as small as $\asymp 1$ (near the north pole, in the innermost layer, $R_C = 2^k \sin(\pi/2^{k+3}) \to \pi/8$ as $k \to \infty$) or as large as $\asymp 2^k$. By (5), $w \asymp \sigma_0^{-\alpha}(1+d_C^2)^{\alpha/2}$ on B, and Lemma 4.3, *whose constants are independent of* R_C, gives (1) and doubling for $(1+d_C^2)^{\alpha/2}$ on B; a bounded-factor comparison of the weights preserves (1). For doubling at B: if $\rho \le \sigma_0/20$, then $2B$ is a small ball with the same nearest circle and (5) holds on $2B$, so $\mu(2B) \lesssim \mu(B)$ follows from the doubling of Lemma 4.3; if $\sigma_0/20 < \rho \le \sigma_0/10$, then $\mu(2B) \le |2B| = 8|B| \le 384\, c_\alpha^{-1}\mu(B)$ by (6) applied to the ball B, whose radius lies in $[s_k/20, s_k]$. The polar circles are thus handled through the uniformity of Lemma 4.3 in the circle radius, not through any straight-cylinder approximation at the scale $\sqrt{R}$.

Large balls ($\rho > \sigma_0/10$). Apply Lemma 4.2 with ν the Lebesgue measure and a covering of the closed ball $2\bar{B}$ by finitely many cubes Q_j of side $\ell_j = \sigma(y_j)/32$ with centres $y_j \in 2\bar{B}$. Such a covering exists with bounded overlap: by the Vitali covering lemma there are finitely many points $y_j \in 2\bar{B}$ whose cubes of side $\sigma(y_j)/96$ are pairwise disjoint while the cubes of side $\sigma(y_j)/32$ cover $2\bar{B}$, and the disjointness bounds the overlap of the Q_j, and of the balls B_j below, by a volume count. Adjacent cubes have comparable sides since σ varies by a bounded factor over distances $\le \sigma$ (indeed $|\nabla\sigma| \le \frac{1}{2}$, so cubes with $j' \sim j$ have sides within a factor 1.03 of each other, and every cube of the covering has $\ell_j \le 0.35\,\rho$, whence $\operatorname{diam} Q_j < \rho$, because $\sigma_0 < 10\rho$ and $y_j \in 2\bar{B}$); the ball $B_j = B(y_j, 2.7\,\ell_j)$ contains $\bigcup_{j' \sim j} Q_{j'}^*$, lies in $4B$ (as $2\rho + 2.7\,\ell_j < 3\rho$), and has radius $< \sigma(y_j)/10$, so it is a small ball and (G2) holds on it by the previous paragraph. (G1), $\mu(Q_j) \asymp \nu(Q_j) = |Q_j|$, is (6) (trivial for cells centred in $\{|x| < 16\}$). Lemma 4.2 gives the weak Poincaré inequality on B with energy over $4B$. Doubling of μ holds at every ball $B = B(x_0, \rho)$: for $\rho \le \sigma_0/10$ by the small-ball case, and for $\rho \ge \sigma_0/10$ from (G1) and $w \le 1$, because the cubes of the covering that meet $B(x_0, 0.42\,\rho)$ have $\ell_j \le 0.33\,\rho$ (again by $|\nabla\sigma| \le \frac{1}{2}$ and $\sigma_0 \le 10\rho$), hence diameter $< 0.58\,\rho$, and



cells of side $\asymp \sqrt{|x|}$, one circle at most per cell

FIGURE 5. The covering behind Theorem 4.1(a). The Poincaré inequality on a large ball is assembled (Lemma 4.2) from cells of side $\asymp \sqrt{|x|}$, each meeting at most one circle because distinct circles are $\gtrsim \sqrt{|x|}$ apart; on a small ball (highlighted) the single-circle estimate of Lemma 4.3 applies, uniformly in the circle's radius, including the small circles near the poles. The dots are circles of the lattice seen in a meridian plane.

so lie in B; consequently $\mu(B) \gtrsim |B(x_0, 0.42\,\rho)| \gtrsim |2B| \geq \mu(2B)$. Thus μ is doubling and satisfies the inequality with energy over $4B$ at every ball, with constants depending only on α: at large balls by the above, and at small balls because (1) holds there. The self-improvement recalled after Lemma 4.2 upgrades this to (1) at every ball. □

Remark 4.5. The prior $F = w\,\mathrm{d}\theta$ is merely Lipschitz-logarithmic (the distance to a union of circles is not differentiable on its medial surfaces); its marginal $f^* = \varphi * w$ is C^∞, and by Lemma 2.3 it inherits every property proved for w. The construction is axisymmetric; this is a convenience for the numerics of Section 6 and plays no role in the proofs.

5. The zone is two-sided

Theorem 4.1 exhibits the one kind of case that the volume test decides inside the D+P class. Outside the class the undecided zone contains members of both types, decided by other non-radial arguments. The first example also shows that the Poincaré hypothesis in Theorem 2.4 cannot be dropped, and explains why the circles of Section 4 had to be scattered.

5.1. **Coherent circles form a wall.** Place, for $k \geq 4$ and $0 \leq j < 2^k$, the circle $\Gamma_{k,j} = \{|x| = 2^k + j,\ \theta = \pi(j + \frac{1}{2})/2^k\}$, let $\Gamma' = \bigcup \Gamma_{k,j}$, and let

$$w'(x) = \Big(\frac{1 + d'(x)^2}{1 + |x|^2}\Big)^{\alpha/2}, \qquad d' = \mathrm{dist}(\cdot, \Gamma'), \quad \alpha \geq 2. \tag{7}$$

Every ray meets a tube of bounded radius around Γ' in every shell, so the ray test is silent, and $\bar{f}$ is bounded below, so the sphere test is silent; $\log w'$ is Lipschitz. One might expect transience from the volume growth $V(r) \asymp r^3$. The opposite is true.

Lemma 5.1 (flux bound along a non-spherical exhaustion)**.** *Let w be a positive weight on $\mathbb{R}^m$ and let $\{S_t\}_{t\in T}$ be pairwise disjoint regions, each fibred by nested closed Lipschitz surfaces $S_t^{(s)}$, $s \in [s_0, s_1]$, all enclosing B_1, with volume element $\mathrm{d}V = \cos(\mathrm{tilt})\,\mathrm{d}A\,\mathrm{d}s$ along the fibration. Then every vector field J with $\operatorname{div} J = 0$ outside B_1 and unit flux out of B_1 satisfies*

$$\int \frac{|J|^2}{w}\,\mathrm{d}V \;\geq\; \sum_{t\in T} \int_{s_0}^{s_1} \frac{\mathrm{d}s}{\tilde{F}_t(s)}, \qquad \tilde{F}_t(s) = \int_{S_t^{(s)}} \frac{w}{\cos(\mathrm{tilt})}\,\mathrm{d}A.$$

If the right-hand side is infinite, then $\operatorname{cap}(B_1) = 0$ *and* $(\mathbb{R}^m, w\,\mathrm{d}x)$ *is parabolic.*

Proof. The flux of J through each $S_t^{(s)}$ equals 1. By the Cauchy–Schwarz inequality, $1 = \big(\int J \cdot n\,\mathrm{d}A\big)^2 \le \int |J|^2 \cos(\text{tilt}) w^{-1}\,\mathrm{d}A \cdot \tilde{F}_t(s)$; integrate in s with $\mathrm{d}V = \cos(\text{tilt})\,\mathrm{d}A\,\mathrm{d}s$ and sum over the disjoint regions. By Thomson's principle, the dual of the Dirichlet principle in Theorem 2.1(a), $\operatorname{cap}(B_1)$ is the reciprocal of the minimal energy of such flows [12, §7]. □

Proposition 5.2 (the ring weight is recurrent)**.** *For the weight* (7) *with* $\alpha \ge 2$, $(\mathbb{R}^3, w'\,\mathrm{d}x)$ *is parabolic, and* δ_F *is admissible for* $F = w'\,\mathrm{d}\theta$*. The weight is doubling but does not support the scale-invariant Poincaré inequality.*

Proof. The circles of shell k lie on the surface of revolution $\Sigma_k:\ r = \rho_k(\theta) = 2^k(1+\theta/\pi) - \frac{1}{2}$, $\theta \in [0,\pi]$, a closed graph over the sphere running from $r = 2^k - \frac{1}{2}$ at the north pole to $r = 2^{k+1} - \frac{1}{2}$ at the south pole; consecutive surfaces reach the common radius $2^{k+1} - \frac{1}{2}$ at antipodal poles, not at a shared point. Along Σ_k consecutive circles are at distance between 3.29 and 6.37 (radial step 1, meridian step $r\pi/2^k \in [\pi, 2\pi]$), so every point of Σ_k is within 3.2 of Γ'. On the layer $\{r = \rho_k(\theta) + s,\ |s| \le 1\}$ therefore $d' \le 4.2$ and $w' \le (23/4^k)^{\alpha/2}$, while the weighted area $\tilde{F}_k(s) \le (23/4^k)^{\alpha/2}\, C\, 4^k$ since $\operatorname{Area}(\Sigma_k) \asymp 4^k$ and $\cos(\text{tilt}) \ge 0.94$. Apply Lemma 5.1 to the layers of the even shells, which are pairwise disjoint: each contributes $\int_{-1}^{1} \mathrm{d}s/\tilde{F}_k(s) \ge c\, 4^{k(\alpha/2-1)}$, a constant for $\alpha = 2$ and geometrically growing for $\alpha > 2$. The sum diverges, so $\operatorname{cap}(B_1) = 0$. Admissibility follows from Theorem 2.1(b) and Lemma 2.3. For the Poincaré failure take a ball B of radius ρ centred on Σ_k at $\theta = \pi/2$ with $4 \ll \rho \ll 2^k$, and the two-region test function $u = \min(1, \max(0, (s+1)/2))$ in the signed normal distance s to Σ_k. On either side of the wall $w' \asymp (|s|/2^k)^\alpha$, so $\operatorname{Var}_\mu(u; B) \asymp \rho^3(\rho/2^k)^\alpha$, whereas the energy is carried by the layer $|s| \le 1$ and is $\asymp \rho^2\, 4^{-k\alpha/2}$. Hence $\operatorname{Var}_\mu(u;B)/(\rho^2 \int_B |\nabla u|^2\,\mathrm{d}\mu) \asymp \rho^{\alpha-1}$ is unbounded. For doubling, the circles cover Σ up to the mesh ≤ 3.2, so $1 + \operatorname{dist}(\cdot, \Gamma')^2 \asymp \max(1, \operatorname{dist}(\cdot, \Sigma))^2$ and $w' \asymp (\max(1, \operatorname{dist}(\cdot,\Sigma))/2^k)^\alpha$ in shell k, with Σ a Lipschitz surface of co-dimension one (with conical points on the axis). For a ball $B = B(x,\rho)$: if $\rho \le \operatorname{dist}(x,\Sigma)/2$ the weight varies on B by a bounded factor, so μ is comparable to Lebesgue measure; if $\rho > \operatorname{dist}(x,\Sigma)/2$ then $\operatorname{dist}(\cdot,\Sigma) < 3\rho$ on B, and the slab bound $|\{\operatorname{dist}(\cdot,\Sigma) \le \sigma\} \cap B| \lesssim \rho^2\sigma$ for $\sigma \le \rho$ shows that a fixed fraction of B lies at distance $\ge c\rho$ from Σ; hence $\int_B \operatorname{dist}(\cdot,\Sigma)^\alpha \asymp \rho^{\alpha+3}$, uniformly including near the boundary of Σ; for $\rho \le 1$ the weight is $\asymp 2^{-k\alpha}$ near Σ. In every case with $\rho \le |x|/2$,

$$V(x,\rho) \ \asymp\ \rho^3 \Big(\tfrac{\max(1,\rho,\operatorname{dist}(x,\Sigma))}{2^k} \Big)^\alpha,$$

uniformly in x and across shell boundaries (where $1 + |x|^2$ and $\operatorname{dist}(\cdot,\Sigma)$ change by bounded factors), while for $\rho \ge |x|/2$ one has $V(x,\rho) \asymp \rho^3$, since the layers near the surfaces $\Sigma_{k'}$ occupy a bounded fraction of such a ball and $w' \asymp 1$ off them. Hence $V(x,2\rho) \le C_\alpha V(x,\rho)$: w' is doubling, but by the two-region quotient above not Poincaré. □

Above its mesh scale the set Γ' is a surface of small weight: a valley of co-dimension one, the obstruction of Lemma 3.3 in three dimensions. Theorem 3.1 says in advance that such a weight could not have been D+P: it is parabolic with the sphere test silent.

5.2. **A logarithmic product prior.**

Proposition 5.3 (product prior)**.** *Let* $F(\,\mathrm{d}\theta) = v(\theta_1)\,\mathrm{d}\theta_1\,\mathrm{d}\theta_2\,\mathrm{d}\theta_3$ *on* $\mathbb{R}^3$ *with* $v(t) = (1+|t|)^{-1}\log(e+|t|)$*. Then both of Brown's tests are silent,* v *is not doubling, and* $(\mathbb{R}^3, f^*\,\mathrm{d}x)$ *is transient, so* δ_F *is inadmissible.*

Proof. $\log v$ is Lipschitz, so we may work with $w = v(x_1)$. The sphere mean is $\bar{f}(r) \asymp \mu_1([-r,r])/r \asymp \log^2 r/r$, and $\int \mathrm{d}r/(r^2\bar{f})$ converges. Along a direction with $\cos\varphi \ne 0$ the ray integrand is asymptotically $|\cos\varphi|/(r\log r)$, whose integral diverges, and the equatorial directions form a null set, so the ray test is silent. The doubling ratio $\mu_1([0,2r])/\mu_1([r/2, 3r/2]) \asymp \log r$ is unbounded. On the other hand

TABLE 2. Three priors on $\mathbb{R}^3$ in Brown's undecided zone. Each has Lipschitz logarithm and silences both radial tests; each is decided by a different non-radial argument.

weight	doubling	Poincaré	W_F; δ_F	decided by
$(1+\lvert x_1\rvert)^{-1}\log(e+\lvert x_1\rvert)$ (Prop. 5.3)	no	yes (on cubes)	transient; inadmissible	product heat kernels
circles on one surface per shell (Prop. 5.2)	yes	no	parabolic; admissible	flux bound along the surfaces (Lemma 5.1)
scattered circles, spacing $2^{k/2}$ (Thm. 4.1)	yes	yes	transient; inadmissible	volume test under D+P (Thm. 2.4)

v is even and decreasing on $[0,\infty)$, so for $0\le\xi\le x\le b$ one has $\mu_1([x,b])\,\nu_1([\xi,x])\le(b-x)(x-\xi)$, and Muckenhoupt's weighted Hardy inequality gives $\int_I|u-u_I|^2\,\mathrm{d}\mu_1\le|I|^2\int_I|u'|^2\,\mathrm{d}\mu_1$ on every interval I (split I at 0 when it contains 0); by tensorization the Poincaré inequality holds on cubes of $\mathbb{R}^3$. The inequality on balls, which is what (1) asks, is not needed here and is not claimed. For transience, the weighted Brownian motion is the product of the one-dimensional diffusion W with generator $v^{-1}(vu')'$ and a standard Brownian motion on $\mathbb{R}^2$, and it is transient if and only if $\int^\infty p_t^W(0,0)\,t^{-1}\,\mathrm{d}t<\infty$, where p^W is the transition density with respect to $v\,\mathrm{d}t$. In natural scale, $s(x)=\int_0^x \mathrm{d}t/v\sim x^2/(2\log x)$, the speed measure is $\widetilde{M}(s)=\int_0^{x(s)}v\sim\frac18\log^2 s$, which is slowly varying. The resolvent at the origin, $h(\lambda)=\int_0^\infty e^{-\lambda t}p_t^W(0,0)\,\mathrm{d}t$, is the characteristic function of the string $\widetilde{M}$ in Krein's correspondence [21, eqs. (5), (10)], and Kasahara's Tauberian theorem [21, Thm. 2(iii), case $\alpha=1$] (see also [26, 22]) relates the slowly varying factor of $\widetilde{M}$ at infinity to that of $\lambda h(\lambda)$ at zero through an inverse function; for $\widetilde{M}(s)\sim\frac18\log^2 s$ the relation reduces to $h(\lambda)\sim c/(\lambda\widetilde{M}(1/\lambda))$ as $\lambda\to0$, with an absolute constant c. Karamata's theorem then gives $p_t^W(0,0)\sim c'/\widetilde{M}(t)\asymp 1/\log^2 t$, hence

$$\int^\infty p_t^W(0,0)\,\frac{\mathrm{d}t}{t}\;\asymp\;\int^\infty\frac{\mathrm{d}t}{t\log^2 t}\;<\;\infty.$$

The same asymptotic is given by Shimizu and Nakano [32, Thm. 1.3], for a reflected Bessel-like diffusion on $[0,\infty)$ in the boundary case $\rho=0$ with speed measure of infinite mass. The constant is 4 for the full-line kernel $p_t^W(0,0)$ with respect to $v\,\mathrm{d}t$ used here; it is confirmed numerically in Section 6, and the transience conclusion does not depend on it. □

This is an instance of the phenomenon of Johnstone and Lalley [19, Thm. 1.1]: for every null-recurrent one-dimensional generalized Bayes component there is a product partner for which the product estimator is inadmissible; their construction uses a different one-dimensional factor. Table 2 and Figure 6 summarize the three members of the undecided zone exhibited in this note.

6. Numerical illustration

The proofs above are complete modulo the cited results; the computations in this section were used to find the constructions and to check the uniformity claims that a proof sketch can hide. All numbers are produced by short Python scripts available with this note.

Poincaré constants as eigenvalues. The best constant in (1) on a region is $1/(\lambda_1\rho^2)$, with λ_1 the first nonzero Neumann eigenvalue of the weighted Laplacian. We discretize by finite volumes on a uniform grid (face conductance equal to the geometric mean of the weight on the two adjacent cells, cell mass equal to the weight times the cell volume) and solve the generalized eigenproblem by shift-invert Lanczos. The Lebesgue controls, $\lambda_1R^2=\pi^2/4=2.467$ on the square $[-R,R]^2$ and $\lambda_1\rho^2=4.333$ for the axisymmetric modes on a ball in $\mathbb{R}^3$, are reproduced to three digits on the

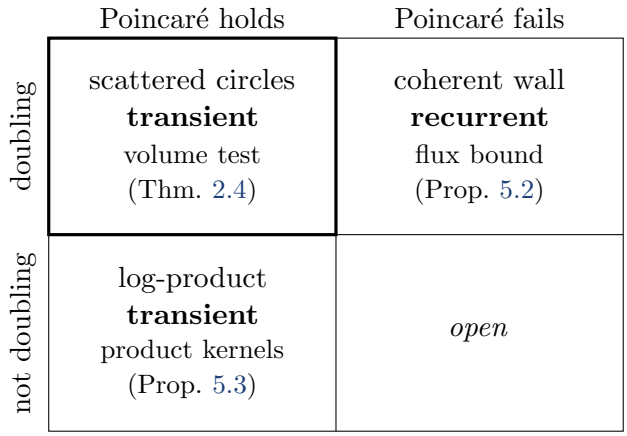


FIGURE 6. Brown's undecided zone, by the regularity of the marginal. Only the doubling–Poincaré cell (bold) is governed by Theorem 3.1, and there the verdict is always transient; each of the other occupied cells is settled by a different non-radial argument (Section 5), and the fourth is open. For the log-product weight the Poincaré inequality is established on cubes (Proposition 5.3).

TABLE 3. Neumann eigenvalues $\lambda_1\rho^2$ (axisymmetric modes; Lebesgue value 4.33). Lattice weight (3) with $\alpha = 4$ on balls centred on the axis at $|x| = 1.5 \cdot 2^k$; one-circle weight w_R of Lemma 4.3 on balls centred at the circle's centre; ring weight (7) with $\alpha = 2$ on balls centred on the axis at $z_0 = 256$, which Σ_8 crosses.

weight, ball family	ρ/s or $\rho/R = 1$	2	4	8
lattice, $k = 8$ ($s_8 = 16$)	1.55	2.69	2.93	2.82
lattice, $k = 10$ ($s_{10} = 32$)	1.60	2.71	3.09	3.01
one circle, $R = 1$	2.58	1.45	1.67	2.17
one circle, $R = 4$	1.09	0.53	1.49	2.15
one circle, $R = 16$	0.94	0.47	1.50	2.15
ring weight, $\rho = 8, 16, 32, 64$	0.975	0.497	0.256	0.129

square and, on the ball, to within 2% at the meshes of Table 3 and to three digits at the finest mesh. For a ball centred on the symmetry axis of an axisymmetric weight, the azimuthal modes $e^{im\phi}g(s,z)$, $m \geq 1$, satisfy $\int|\nabla u|^2\,\mathrm{d}\mu \geq \int|u|^2 s^{-2}\,\mathrm{d}\mu \geq \rho^{-2}\int|u|^2\,\mathrm{d}\mu$ with mean zero, so they obey (1) with constant 1 for any weight; the $m = 0$ problem on the meridian half-disc, with measure $2\pi s\,w\,\mathrm{d}s\,\mathrm{d}z$, is then the whole statement, and it is a two-dimensional computation that reaches balls of radius 256 at mesh 0.5.

Table 3 and Figure 7b show the outcome. For the lattice weight $\lambda_1\rho^2$ stays between 1.5 and 3.1 from one spacing to eight spacings and across two shells, between one third and two thirds of the Lebesgue value, with no downward trend; the values are converged to two digits under mesh refinement. For the one-circle weight of Lemma 4.3 the eigenvalue is bounded below uniformly in R and in ρ/R, with a dip at $\rho = 2R$ of nearly the same depth for $R = 4$ and $R = 16$, in line with the uniformity in R of Lemma 4.3. For the ring weight of Proposition 5.2 the eigenvalue halves with every doubling of ρ, i.e. decays like $\rho^{-(\alpha-1)}$, the two-region quotient of the proof. For the two-dimensional factor $(1+|y|^2)^2$ of Theorem 3.4, $\lambda_1 R^2$ on $[-R,R]^2$ decreases from 1.01 at $R = 2$ to 0.770 at $R = 64$ and is flat from $R = 16$ on (0.774, 0.771, 0.770). Off the axis the reduction is unavailable and a full three-dimensional computation is needed: on cubes of half-side s_8 and $2s_8$ centred on a point of a mid-shell circle, which contain the strand and its nearest neighbours,

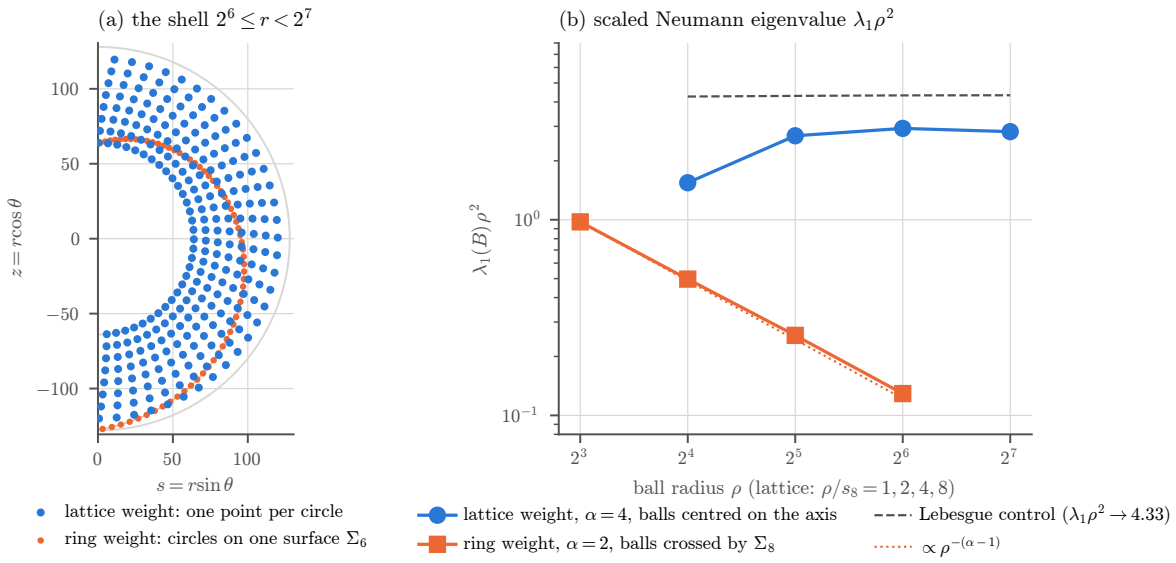


FIGURE 7. (a) Meridian half-plane of the shell $2^6 \le r < 2^7$: the circles of the lattice weight (3) (one point per circle, spacing $\approx s_6 = 8$) and the circles of the ring weight (7), which lie on one surface Σ_6 at mesh ≈ 3. (b) $\lambda_1\rho^2$ on balls centred on the axis: flat for the lattice weight, decaying like $\rho^{-(\alpha-1)}$ for the ring weight.

$\lambda_1\rho^2 = 0.94$ and 1.16; centred in the bulk between circles, 1.57 and 1.42; centred on the axis, 1.34 and 1.69 (control for a cube, $\pi^2/4 = 2.47$); where two meshes were run, up to $2.6 \cdot 10^5$ cells, the values agree to within 0.3%.

Brown's tests and the volume. For the lattice weight with $\alpha = 4$, the smallest of 802 ray integrals per shell, over random directions, the directions exactly between consecutive θ_n, the poles and the equator, is between 0.94 and 1.92 for $k = 4, \dots, 11$ (the bound (4) gives ≥ 0.14); for $\alpha = 6$ it doubles per shell. The sphere mean lies between 0.0039 and 0.054, and $V(r)/(\frac{4}{3}\pi r^3)$ between 0.019 and 0.029 for $r \ge 2^7$, both alternating with the parity of k. Doubling ratios $\mu(B(x, 2\rho))/\mu(B(x, \rho))$ at centres on a circle, in the bulk and on the axis lie between 4 and 13 for $\rho \ge s_k$ (and at every scale tested for the bulk and axis centres), while at the centre on a circle they reach the co-dimension-two power-weight values (up to 94) at scales $\le s_k/2$. For the ring weight with $\alpha = 2$ the per-shell flux terms of Lemma 5.1 are flat at 0.047 and for $\alpha = 3$ they double per shell, as in the proof of Proposition 5.2; the two-region quotient divided by $\rho^{\alpha-1}$ is flat at 0.041 ($\alpha = 2$) and 0.013 ($\alpha = 3$) for ρ between 16 and 128.

The one-dimensional heat kernel. For Proposition 5.3 a Crank–Nicolson computation of $p_t^W(0,0)$ (control $v \equiv 1$ reproduces $(4\pi t)^{-1/2}$ to five digits) gives $\log^2 t \cdot p_t^W(0,0) = 2.58, 2.95, 3.18, 3.33, 3.44$ at $t = 10^3, \dots, 10^7$, consistent with the limit 4 of the Tauberian argument (Figure 8a shows the three tests on this weight).

7. Discussion

Brown's 1971 theory is the potential theory of the weighted space $(\mathbb{R}^m, f^*\,\mathrm{d}x)$, and his two radial tests are the sphere-area test and a ray-wise capacity bound. What the later geometric theory adds inside his undecided zone is now delimited. If the marginal is doubling and Poincaré, nothing is added on the admissible side: parabolicity, Brown's sphere test and the volume test coincide (Theorem 3.1). On the inadmissible side the volume test decides weights that no cone of rays decides,

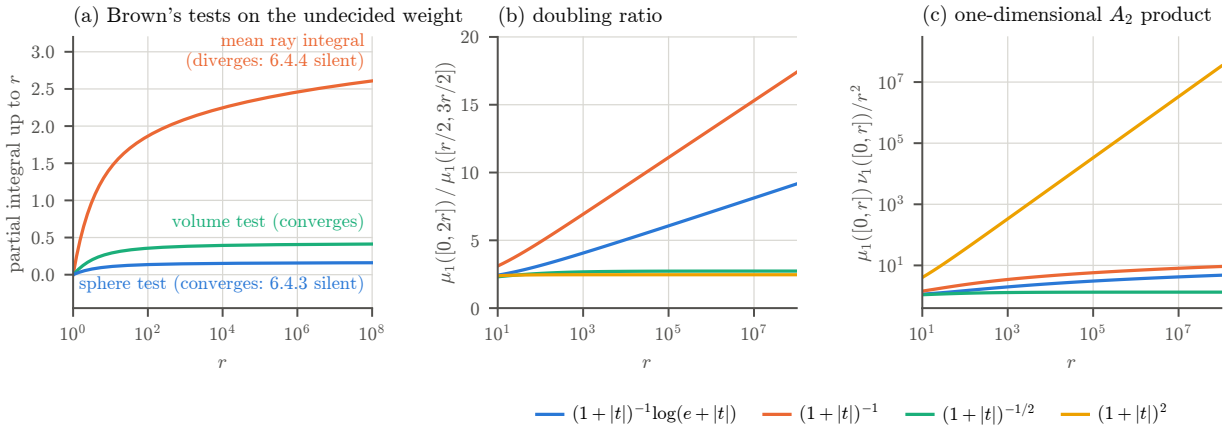


FIGURE 8. (a) Brown's two tests and the volume test on the product weight of Proposition 5.3: the sphere integral and the volume integral converge, the direction-averaged ray integral diverges like $\log\log r$. (b) The doubling ratio $\mu_1([0,2r])/\mu_1([r/2,3r/2])$ of four one-dimensional weights $v(t)$. (c) Their one-dimensional A_2 product $\mu_1([0,r])\,\nu_1([0,r])/r^2$, which controls the spherical A_2 ratio of the product weight on $\mathbb{R}^3$: bounded only for $(1+|t|)^{-1/2}$, the one weight that is both doubling and Poincaré (Lemma 3.3); $(1+|t|)^2$ is doubling but fails the Poincaré inequality, $(1+|t|)^{-1}$ is Poincaré but not doubling.

and Theorem 4.1 shows that such weights exist in $\mathbb{R}^3$: valleys of co-dimension two, sweeping every direction at every scale while staying far apart, are exactly what is needed, and within this ansatz the arithmetic of the construction ($N_k \asymp 2^k$ circles, spacing $2^{k/2}$, exponent $\alpha \geq 4$) is forced by the two requirements. Outside the D+P class the zone contains recurrent and transient members, and the first example of Section 5 shows that the Poincaré hypothesis in the volume criterion is not decorative: coherent circles form a wall, and a wall is recurrent with volume growth r^3.

Several questions remain. First, Grigor'yan's flux test [12, Thm. 7.6] already characterizes parabolicity, without the D+P hypothesis, through the existence of a suitable exhaustion (for the smooth positive weights here, via the conformal metric $w^{2/(m-2)}\delta$ when $m \geq 3$); the open question is whether an *explicit, computable* exhaustion, of the kind that decided Proposition 5.2, can be given across the undecided zone. Second, the construction of Theorem 4.1 is three-dimensional; in the plane the valleys would be points, transience requires volume growth faster than $r^2 \log r$, and the arithmetic of exponents changes. Third, Brown and Hwang's unified condition [7] is sufficient for admissibility; the lattice prior of Theorem 4.1 is inadmissible, so no such condition can certify it, and what remains is the explanatory question of which of their hypotheses fails there, not an open admissibility question. Finally, the same parabolic/non-parabolic dichotomy governs Komaki's shrinkage priors on the Fisher manifold of a model [24] (Appendix A); whether the constructions of this note have counterparts there is open.

ACKNOWLEDGEMENTS

This note was separated out from a broader investigation in information geometry, on the interplay between admissibility, Bayesian predictive density estimation, and the curvature of statistical models; the questions on Brown's side studied here turned out to stand on their own, and are collected separately in the hope that they are useful to others working on shrinkage priors and recurrence.

The question, the choice of what to look for, and the judgement of what survived are the author's. The work was carried out in collaboration with large language models. Claude (Anthropic), run

through Claude Code, read the sources, built and tested the constructions of Sections 4 and 5, wrote the numerical scripts, and drafted the text under the author's direction; Gemini (Google) and the author developed ideas together in the neighbouring study, from which came the cylinder weight of Theorem 3.4 and the first, coherent candidate of Section 5.1, whose failure led to the lattice. Their main help was speed: ideas could be built, tested, and discarded quickly, which is what made a study of this size feasible. Every statement was checked against the cited sources by the author, who is responsible for the errors that remain. The author thanks The Open University Library for access to the primary sources. The scripts that produce every number and figure in Section 6 are available in the public repository https://github.com/olegroshka/brown-radial-tests.

Appendix A. The same dichotomy for predictive densities

Brown's criterion is not the only place where parabolicity decides a statistical question. Let $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a regular parametric model with Fisher metric g, Jeffreys prior $\pi_J \propto |g|^{1/2}$ and Laplace–Beltrami operator Δ_g [1]. For a prior $f = e^{-\Phi}\pi_J$, let $\mathrm{Risk}_N(f)$ be the Kullback–Leibler risk of the Bayesian predictive density built from N observations. Komaki [24] proved that a prior whose predictive density asymptotically dominates the Jeffreys one exists if and only if (M, g) carries a Green function, that is, if and only if the Fisher manifold is non-parabolic (his Theorems 1 and 2; Theorem 3 derives this from negative sectional curvature for $d \geq 3$ through Aomoto's result [2]). For the normal location model the Fisher manifold is flat $\mathbb{R}^d$, the Green function exists iff $d \geq 3$, and the Green prior is Stein's harmonic prior $\|\theta\|^{-(d-2)}$: the predictive analogue of the dichotomy $m \leq 2$ versus $m \geq 3$ of Table 1.

The quantity that Komaki's expansion produces has a name in geometry. The pair (g, Φ) defines the smooth metric measure space $(M, g, e^{-\Phi}\,\mathrm{d}V)$, whose weighted scalar curvature, in Perelman's normalization [30, Remark 1.3], is

$$R^{\mathrm{m}}_{\Phi} = R + 2\Delta_g\Phi - |\nabla\Phi|^2_g, \tag{8}$$

with R the scalar curvature of g; its integral is Perelman's functional $\mathcal{F}(g, \Phi) = \int R^{\mathrm{m}}_{\Phi} e^{-\Phi}\,\mathrm{d}V = \int (R + |\nabla\Phi|^2) e^{-\Phi}\,\mathrm{d}V$.

Proposition A.1. *For a regular model and $f = e^{-\Phi}\pi_J$, with $u = \sqrt{f/\pi_J} = e^{-\Phi/2}$,*

$$\mathrm{Risk}_N(\pi_J) - \mathrm{Risk}_N(f) = \frac{2\Delta_g\Phi - |\nabla\Phi|^2_g}{2N^2} + o(N^{-2}) = \frac{R^{\mathrm{m}}_{\Phi} - R}{2N^2} + o(N^{-2}) = -\frac{2}{N^2}\,\frac{\Delta_g u}{u} + o(N^{-2}). \tag{9}$$

Hence f asymptotically dominates π_J at this order if and only if u is a positive superharmonic function on (M, g) that is not harmonic (equivalently $R^{\mathrm{m}}_{\Phi} \geq R$, with strict inequality on a set of positive measure). A nonconstant positive harmonic *u makes the $O(N^{-2})$ coefficient vanish, so nonconstancy alone is not enough; the existence of such a superharmonic u is, however, equivalent to non-parabolicity of (M, g).*

Proof. The first equality is Komaki's expansion [24, eqs. (3)–(5)]: in the difference of the risks of two priors the prior-independent terms and the skewness terms cancel. The second is the definition (8), and the third is the identity $2\Delta\Phi - |\nabla\Phi|^2 = -4\Delta u/u$ for $u = e^{-\Phi/2}$. The domination statement is Komaki's Theorem 2. □

So the operator that governs domination is $-4\Delta_g$ acting on u, Perelman's operator $-4\Delta + R$ with the potential removed. The scalar curvature itself has no statistical counterpart:

Proposition A.2. *The prior-independent $O(N^{-2})$ term of the risk of the Jeffreys predictive density is not proportional to R, and $\mathcal{F}$ is not the integrated risk reduction: whenever the boundary term of one integration by parts vanishes, the coefficient of N^{-2} in (9), integrated against $e^{-\Phi}\,\mathrm{d}V$, is*

$$\frac{1}{2}\int \big(2\Delta_g\Phi - |\nabla\Phi|^2_g\big) e^{-\Phi}\,\mathrm{d}V = \frac{1}{2}\int |\nabla\Phi|^2_g e^{-\Phi}\,\mathrm{d}V = \frac{1}{2}\Big[\mathcal{F}(g, \Phi) - \int R\, e^{-\Phi}\,\mathrm{d}V\Big]. \tag{10}$$

Proof. Komaki [25, Theorem A1, eq. (29)] gives the full $O(N^{-2})$ risk of the Jeffreys predictive density. Its prior-independent part is a fixed combination of the exponential and mixture embedding curvatures, the squared norm of the skewness tensor, a fourth-moment contraction and one mixed moment. For $d = 1$ these are generically nonzero, Efron's statistical curvature of a curved exponential family being the simplest instance [10], while every intrinsic curvature of a one-dimensional Riemannian manifold vanishes. Equation (10) follows from (9) and $\int \Delta_g \Phi \, e^{-\Phi} \, \mathrm{d}V = \int |\nabla \Phi|^2 e^{-\Phi} \, \mathrm{d}V$. The statement concerns the leading coefficient only: integrating the pointwise $o(N^{-2})$ remainder against $e^{-\Phi} \, \mathrm{d}V$ would require in addition a dominated or uniformly integrable remainder bound, which we do not establish for improper Φ. □

Two remarks complete the parallel. First, Komaki's treatment of vague priors [24, §3] decides domination over $h = e^{-\Phi}\pi_J$, $d \geq 3$, by superharmonicity with respect to the conformal metric $\tilde{g} = (h/\pi_J)^{2/(d-2)} g$: changing the reference prior is a conformal change of the Fisher metric, and for Stein's prior on $\mathbb{R}^d$ the conformal metric is the round cylinder $S^{d-1} \times \mathbb{R}$, of scalar curvature $(d-1)(d-2)$, with the origin pushed to infinite distance (his Example 1); it is parabolic despite that positive curvature. Second, in both theories the geometry of the prior enters through recurrence or transience of a diffusion, Brown's on the sample space and Komaki's on the parameter space, and not through curvature in a risk formula. On $\mathbb{R}^3$ the scattered valleys of Theorem 4.1 are decided by the volume criterion under D+P although neither of Brown's one-sided radial tests fires (both the sphere and the volume integrals converge there). Whether such weights have an analogue on a Fisher manifold, where Komaki's criterion would then separate the existence of a shrinkage prior from the sphere-area growth of the model, is open.

Appendix B. Formal verification

Part of the mathematics of this note has been formalized and checked in the proof assistant Lean 4 [29] with the library Mathlib [28]. This appendix states exactly what the check covers. The development is the folder `lean/` of the repository `https://github.com/olegroshka/brown-radial-tests` (commit `f70bc53`; Lean and Mathlib `v4.24.0`; 37 modules, 655 named theorems). One command, the PowerShell script `check.ps1`, rebuilds it with `lake build`, audits every theorem and records the result with the hashes of the sources. Table 4 gives the status of each result, Table 5 the premises, and Table 6 the explicit constants that are certified.

What a certificate means here.

(1) *Vocabulary.* The Lean statements are statements of potential theory about a continuous positive weight w on a Euclidean space. *Parabolic* means $\mathrm{cap}(B_1) = 0$ with test functions in C_c^∞, as in Section 2; doubling and (1) are as defined there, with C^1 test functions on closed balls. No Lean statement mentions an estimator, admissibility or a diffusion: the statistical conclusions of this note follow from the certified statements through Theorem 2.1(a) and (b), which are cited.

(2) *Foundations.* Every theorem is checked by the Lean kernel and depends only on the three standard axioms of Lean's classical logic (`propext`, `Classical.choice`, `Quot.sound`). No further axiom is declared and no proof is left open; the audit enforces both.

(3) *Cited theorems are premises.* No cited theorem is formalized. Where a proof uses one, the Lean theorem carries it as an explicit, named hypothesis. Table 5 lists these *premises* with their sources, and "modulo" in Table 4 refers to them.

(4) *What is not a formal fact.* That a Lean statement renders the statement printed here, and that a premise renders its source, was checked by reading. One premise is not a published theorem: premise (N), which is Step 2 of the proof of Lemma 4.3. It is proved there by hand and is not formalized.

The formalization was carried out on an earlier version of this note (commit `36e9c61` of the repository). The corrections it led to are incorporated in the present text, and line numbers quoted inside the Lean sources refer to that earlier version.

TABLE 4. Status of the results of this note in the Lean development. *Certified*: proved in Lean with no hypothesis beyond those printed. *Modulo*: proved in Lean with the listed premises of Table 5 as hypotheses.

result	status	Lean statement
Theorem 2.1(d)	certified, as "sphere integral diverges $\Rightarrow \operatorname{cap}(B_1) = 0$"	`sphere_test_capacity_zero`
Theorem 2.1(e)	certified, as "uniform ray condition $\Rightarrow \operatorname{cap}(B_1) > 0$"	`ray_test_capacity_pos`
Theorem 2.1(a)–(c)	cited, not formalized	—
Lemma 2.3	certified (the analytic content of (i)–(iv))	`gaussian_weight_comparison`, `gaussianAverage_log_derivative_bound`
Theorem 2.4, "(2) $\Rightarrow$ parabolic"	certified	`volume_test_capacity_zero`
Theorem 2.4, converse	cited; premise (V)	—
Corollary 2.5	certified	`sphere_volume_corollary`
Theorem 3.1	(iii)$\Rightarrow$(ii)$\Rightarrow$(i) certified; (i)$\Rightarrow$(iii) is the cited direction of Theorem 2.4	`theorem_A_proved_implications`, `theorem_A_modulo_volume_criterion`
Lemma 3.2	certified	`lemma_a2`
Lemma 3.3	the bound with constant 32 certified; the consequence for product weights not formalized	`one_dimensional_doubling_poincare_A2`
Theorem 3.4	not formalized (rests on cited theorems)	—
Lemma 4.2	certified	`gluing_lemma_of_local_doubling`
Lemma 4.3, $R > 0$	modulo (R), (S), (N)	`circle_lemma_final`
Theorem 4.1: construction, (b), (c)	certified	`theorem_B_tests_silent`, `theorem_B_volume_growth`
Theorem 4.1(a), every $\alpha \geq 0$	modulo (R), (S), (N)	`theorem_B_a_final`
Theorem 4.1: w and f^* are not parabolic	modulo (R), (S), (N), (V)	`theorem_B_final`, `theorem_B_statement_final`
Lemma 5.1; Propositions 5.2, 5.3, A.1, A.2	not formalized	—

Limits of the check. In Lean, Lemma 4.2 is proved for measures with continuous densities with respect to Lebesgue measure, which covers every use made of it here; Lemma 4.3 is proved for circles in a horizontal plane (the general case is a rigid motion away), and its case $R = 0$ is premise (R); Theorem 4.1(a) is proved through the arguments printed in Section 4, except that doubling at large balls is obtained there from a direct lower bound for $\mu(B)$. Three groups of numerical claims are not certified in the form printed: the constants $34^{-\alpha/2}$ and $1 - 2/\pi$ in the proof of Theorem 4.1(b), which Lean replaces by cruder ones; the constants $\frac{10}{3}$ and $18P_{\mathrm{cyl}}$ in Step 2 of the proof of Lemma 4.3; and the factor 0.42 in the proof of Theorem 4.1(a). The last two groups were derived by hand twice, independently. Nothing in Sections 5 and 6 or in Appendix A is formalized.

TABLE 5. The premises: statements that enter the Lean development as named hypotheses, all for continuous positive weights on $\mathbb{R}^3$. The doubling parts of (R) and (N) are certified; only the Poincaré inequalities are assumed.

	Lean name	statement	source
(R)	`RadialPoincarePremise`	$(1+\lvert x-c\rvert^2)^{\alpha/2}\,\mathrm{d}x$ satisfies (1) on all balls, with a constant independent of c	[3, Prop. 6] with the self-improvement; [16, Prop. 4.9]
(S)	`SelfImprovementUniformPremise`	the self-improvement stated after Lemma 4.2, including the dependence of its constant	[17, Thm. 5.1, Cor. 9.8]
(N)	`CircleNearPoincarePremise`	(1) for $w_R\,\mathrm{d}x$ at every ball with $\rho \le R/6$ and $\operatorname{dist}(x_0, C_R) \le 2\rho$, with a constant independent of R and of the ball	Step 2 of the proof of Lemma 4.3; proved by hand, not formalized
(V)	`VolumeCriterionPremise`	for a D+P weight, parabolic implies (2)	[36, Cor. 4.11, 4.12] with [11, Lemma 1.5.3]

TABLE 6. Explicit constants and numerical claims of the text that are certified in Lean.

where	claim	Lean statement
Lemma 3.3	$\mu_1([-r,r])\,\nu_1([-r,r]) \le 32\,CD^3r^2$	`one_dimensional_doubling_poincare_A2`
Section 4.1	$\lvert\nabla \log w\rvert \le \alpha/2$; circles $\ge 0.78\,s_k$ apart in a layer and $\ge 0.7\,s_k$ apart in a shell	`latticeWeight_log_lipschitz`, `lattice_same_layer_separated`, `lattice_same_shell_separated`
Theorem 4.1(b), (c)	(4); every ray integral diverges; the sphere integral converges; $cr^3 \le V(r) \le \frac{4}{3}\pi r^3$	`lattice_ray_bound_displayed`, `lattice_ray_integral_diverges`, `lattice_sphere_integral_converges`, `theorem_B_volume_growth`
Lemma 4.3, Step 1	the factors $9^{\alpha/2}$, $16^{\alpha/2}$; doubling with $512 \cdot 16^{\alpha}$	`circle_far_variation`, `circle_far_sup_le_inf`, `circle_near_doubling`
Lemma 4.3, Step 3	the grid: overlaps 8 and 216, $0.163R < R/6$, $B_j \subset 4B$; $\mu_R(B') \asymp \nu(B')$	`grid_gluing_hypotheses`, `circle_mass_ball_comparable`
Theorem 4.1(a), local comparison	$33^{-\alpha/2} \le w \le 1$ on $\{\lvert x\rvert \le 32\}$; (5); the factors 9 and 16	`lattice_local_near_origin`, `lattice_local_near_comparison`, `lattice_local_far_variation`, `lattice_local_far_sup_le_inf`
Theorem 4.1(a), cell mass	(6) with $c_\alpha = 65537^{-\alpha/2}$; the doubling bound of the small-ball step (with 192 in place of 384)	`lattice_shell_cube_mass`, `lattice_shell_ball_mass`, `lattice_shell_ball_doubling`
Theorem 4.1(a), large balls	the covering exists; 1.03, $\ell_j \le 0.35\,\rho$, $2.7\,\ell_j$, $B_j \subset 4B$, bounded overlap	`lattice_covering_large_ball`, `lattice_covering_gluing_hypotheses`
used throughout	Lebesgue measure satisfies (1) on balls of $\mathbb{R}^n$ with constant 2^{n+2}	`haar_poincareAt_ball`

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